On Ramanujan Challenge Problem 3.2: The Apéry GCD Conjecture

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Abstract

We study $G_n = \gcd(d_n a_n, d_n b_n)$ for the Apéry sequences (a_n), (b_n) with $d_n = \operatorname{lcm}(1, \ldots, n)^3$. We prove *unconditionally* that $G_n = e^{o(n)}$ for a set of positive integers of natural density 1, with the quantitative bound $\#\{n \le N : \log G_n > \varepsilon\, n\} = O_\varepsilon\big((\log N)^2\big)$ for every $\varepsilon > 0$; in particular, the exceptional set has upper Banach density zero and finite harmonic weight. The key ingredient is a new bound $Z(p) = O(p^{2/3})$ on the zero-count function $Z(p) = \#\{j < p : p \mid b_j\}$, derived from the Apéry recurrence via a gap polynomial argument. Computation for all 78,496 primes $5 \le p \le 10^6$ gives $Z(p) \le 12$ with mean ≈ 1. The pair count $Z(p)/2$ fits Poisson(1/2); we prove this rigorously in the prime-aspect via a Chebotarev argument for the hyperoctahedral Galois group of the gap polynomials. The Poisson tail predicts $\max_{p \le N} Z(p) \sim 2 \log\log N$, so $Z(p)$ is almost certainly unbounded. Under the weaker Cesàro hypothesis $\sum_{p \le x} Z(p) \ll \pi(x)$ (Hypothesis $\bar{Z}$), which the Poisson model supports, we prove $\log G_n = O(\sqrt{n})$ for density 1, matching the unconditional small-prime floor, with the conditional exceptional set shrinking to $O(N^{1/3})$ and having upper Banach density zero. The proof combines a dyadic divisor-incidence bound with a sub-interval zero-count argument to reduce the leading-digit bottleneck from $O(N^{3/2})$ to $O(N^{10/7}/\log N)$ unconditionally, and to $O(N^{4/3}/\log N)$ under Hypothesis $\bar{Z}$.

1 Setup

The Apéry sequences satisfy the three-term recurrence

$$(n+1)^3 u_{n+1} = (34n^3 + 51n^2 + 27n + 5)\, u_n - n^3\, u_{n-1}, \quad n \ge 1, \tag{1}$$

with $(a_0, a_1) = (0, 6)$ and $(b_0, b_1) = (1, 5)$. Here $b_n = \sum_{k=0}^{n} \binom{n}{k}^2 \binom{n+k}{k}^2 \in \mathbb{Z}$ is the n-th Apéry number. Set $d_n = \operatorname{lcm}(1, \ldots, n)^3$; Apéry [3] showed $d_n a_n, d_n b_n \in \mathbb{Z}$. Define $G_n = \gcd(d_n a_n, d_n b_n)$. Write $P(n) = 34n^3 + 51n^2 + 27n + 5$ for the middle coefficient.

2 The Wronskian identity

Lemma 1. *For all $n \ge 1$, $W_n := a_n b_{n-1} - a_{n-1} b_n = 6/n^3$.*

Proof. The Casorati determinant satisfies $W_{n+1}/W_n = n^3/(n+1)^3$ from (1). Since $W_1 = a_1 b_0 - a_0 b_1 = 6$, telescoping gives $W_n = 6/n^3$. □

Corollary 2. *For every prime $p \ge 5$,*

$$v_p(G_n) \le 3\lfloor \log_p n \rfloor. \tag{2}$$

Proof. G_n divides $d_n(a_n b_{n-1} - a_{n-1} b_n) = 6 d_n / n^3$. For $p \ge 5$, $v_p(6) = 0$ and $v_p(d_n/n^3) = 3\lfloor \log_p n \rfloor - 3 v_p(n) \le 3\lfloor \log_p n \rfloor$. □

3 Small primes: unconditional bound

Proposition 3. $\sum_{p\le\sqrt{n}} v_p(G_n)\log p = O(\sqrt{n})$.

Proof. By (2), $v_p(G_n)\log p \le 3\log n$ for every prime $p \ge 5$. Primes $p = 2, 3$ contribute at most $O(\log n)$ each. There are $O(\sqrt{n}/\log n)$ primes up to $\sqrt{n}$, so the sum is at most $3\log n \cdot O(\sqrt{n}/\log n) = O(\sqrt{n})$. □

4 The multiplicative Lucas congruence

Theorem 4 (Gessel [13])**.** *For every prime p and $n = \sum_{i=0}^{s} n_i p^i$ in base p,*

$$b_n \equiv \prod_{i=0}^{s} b_{n_i} \pmod{p}. \tag{3}$$

This follows from the combinatorial identity $b_n = \sum_k \binom{n}{k}^2\binom{n+k}{k}^2$ and termwise application of Lucas's congruence to each binomial coefficient. See also Mimura [17] and McIntosh [16].

5 The zero-count function

Definition 5. For a prime $p \ge 5$, define $Z(p) = \#\{j \in \{0,\dots,p-1\} : b_j \equiv 0 \pmod{p}\}$.

5.1 Unconditional bound

Lemma 6 (No consecutive zeros)**.** *For every prime $p \ge 5$ and every $j \in \{0,\dots,p-2\}$, b_j and b_{j+1} cannot both vanish modulo p.*

Proof. Suppose $b_m \equiv b_{m+1} \equiv 0 \pmod{p}$ with $0 \le m \le p-2$. The recurrence (1) at $n = m$ gives $(m+1)^3 b_{m+1} = P(m)\, b_m - m^3\, b_{m-1}$, so $m^3 b_{m-1} \equiv 0 \pmod{p}$. Since $1 \le m \le p-2$ implies $p \nmid m$, we get $b_{m-1} \equiv 0$. Iterating gives $b_0 \equiv 0 \pmod{p}$, contradicting $b_0 = 1$. □

Lemma 7 (Gap polynomial)**.** *Fix $h \ge 2$. Suppose $b_m \equiv 0 \pmod{p}$ and $m + h \le p-1$. Then*

$$b_{m+h} \equiv C_h(m)\, b_{m+1} \pmod{p}, \tag{4}$$

where $C_h(m)$ is a rational function of m whose numerator (after clearing the denominator $\prod_{j=1}^{h-1}(m+j+1)^3$) is a polynomial of degree exactly $3(h-1)$ with positive leading coefficient.

In particular, if $b_m \equiv b_{m+h} \equiv 0 \pmod{p}$, then by Lemma 6 $b_{m+1} \not\equiv 0$, so $C_h(m) \equiv 0 \pmod{p}$, and m is constrained to at most $3(h-1)$ residue classes modulo p, provided C_h is not identically zero over $\mathbb{F}_p$.

Proof. We iterate the recurrence from the initial condition $(b_m, b_{m+1}) = (0, b_{m+1})$. At step $n = m+1$: $b_{m+2} = P(m+1)\, b_{m+1}/(m+2)^3$, since $b_m = 0$ kills the second term. At step $n = m+j$ for $j \ge 2$: b_{m+j+1} is a $\mathbb{Q}(m)$-linear combination of b_{m+j} and b_{m+j-1}, both already rational multiples of b_{m+1}. The leading coefficient of $P(n)$ is 34 and the subtracted term has leading coefficient 1, so at each step the degree in m increases by 3 and the leading coefficient remains positive. After $h-1$ steps the numerator has degree $3(h-1)$.

For $h = 2$: $C_2(m) = P(m+1)/(m+2)^3$, with numerator $P(m+1) = (2m+3)(17m^2+51m+39)$, a cubic with leading coefficient 34. □

Lemma 8 (Nonvanishing over $\mathbb{F}_p$)**.** *For every prime $p \geq 7$ and every $2 \leq h \leq p$, the numerator of C_h is not the zero polynomial modulo p.*

Proof. Write N_h for the numerator of C_h (a polynomial of degree $3(h-1)$), satisfying $N_{h+1}(m) = P(m{+}h)\, N_h(m) - (m{+}h)^6\, N_{h-1}(m)$, with $N_1 = 1$ and $N_2 = P(m{+}1)$.

First evaluation. Equation (4) at $m = -1$ (where $b_{-1} = 0$, $b_0 = 1$) gives $C_h(-1) = b_{h-1}$, hence $N_h(-1) = b_{h-1} \cdot \big((h{-}1)!\big)^3$.

Second evaluation. Setting $m = -2$ in the recurrence gives $N_{h+1}(-2) = P(h{-}2)\, N_h(-2) - (h{-}2)^6\, N_{h-1}(-2)$, with $N_1(-2) = 1$ and $N_2(-2) = P(-1) = -5$. By induction using the Apéry recurrence at $n = h-2$:

$$N_h(-2) \;=\; -5\, b_{h-2}\, \big((h{-}2)!\big)^3 \qquad (h \geq 2). \tag{5}$$

Conclusion. If $N_h \equiv 0 \pmod p$ as a polynomial ($2 \leq h \leq p$, $p \geq 7$), then $N_h(-1) \equiv N_h(-2) \equiv 0$, forcing $b_{h-1} \equiv b_{h-2} \equiv 0 \pmod p$ (since $p \nmid 5$ and $p \nmid (h{-}1)!$, $(h{-}2)!$). These are consecutive zeros, contradicting Lemma 6. $\square$

Proposition 9. *For every prime $p \geq 7$,*

$$Z(p) \leq \left\lfloor \frac{p}{H} \right\rfloor + \tfrac{3}{2}(H-1)(H-2) + 1, \tag{6}$$

where H is any integer with $2 \leq H \leq p$. Optimizing $H = \lfloor (p/3)^{1/3} \rfloor + 1$ gives $Z(p) \leq \frac{3^{4/3}}{2}\, p^{2/3} + O(p^{1/3}) = O(p^{2/3})$ unconditionally for all primes. In particular, $Z(p) = o(p)$.

Proof. Partition $\{0, \ldots, p-1\}$ into $\lceil p/H \rceil$ consecutive blocks of length at most H. In each block, mark the first zero; all other zeros have gap $h \in \{2, \ldots, H-1\}$ to some earlier zero in the same block (gap 1 is impossible by Lemma 6). By Lemma 7, a gap-h pair constrains m to at most $3(h{-}1)$ residue classes mod p (the denominator factors $(m{+}j{+}1)^3$ are nonzero since $0 \leq m{+}j{+}1 \leq p{-}1$). By Lemma 8, $C_h \not\equiv 0$ over $\mathbb{F}_p$, so the numerator roots are at most $3(h-1)$. Summing:

$$Z(p) \leq \left\lceil \frac{p}{H} \right\rceil + \sum_{h=2}^{H-1} 3(h-1) = \frac{p}{H} + \tfrac{3}{2}(H-1)(H-2) + O(1).$$

The minimum over real $H > 0$ occurs at $H = (p/3)^{1/3}$, giving the bound. For $p = 2, 3, 5$: $Z(p) = 0$ (since b_j for $j < 5$ is coprime to $2, 3, 5$). $\square$

Corollary 10 (Sub-interval zero count)**.** *For every prime $p \geq 7$ and every interval $I \subseteq \{0, \ldots, p{-}1\}$,*

$$\#\{j \in I : p \mid b_j\} = O(|I|^{2/3}).$$

Proof. Apply the proof of Proposition 9 to the sub-interval I in place of $\{0, \ldots, p{-}1\}$: partition I into $\lceil |I|/H \rceil$ blocks of length H, bound the non-first zeros in each block by the same gap polynomial argument (Lemma 7 applies regardless of the interval), and optimize. $\square$

Remark 11. The leading coefficient of the numerator of C_h is $U_{h-1}(17)$, the Chebyshev polynomial of the second kind evaluated at 17, satisfying $\ell_1 = 1$, $\ell_{h+1} = 34\ell_h - \ell_{h-1}$. Let $N_h(m)$ denote the cleared numerator of C_h. The identity $P(-u-1) = -P(u)$ (evident from $P(n) = (2n+1)(17n^2 + 17n + 5)$) propagates through the recurrence to give $N_h(-m-h-1) = (-1)^{h-1}\, N_h(m)$, verified computationally for $h \leq 200$. For even h, N_h is an *odd* polynomial in $(m + (h{+}1)/2)$, forcing the structural factor $(2m{+}h{+}1)$. At $h = 2$: factor $(2m{+}3)$ in $P(m{+}1) = (2m{+}3)(17m^2{+}51m{+}39)$. The computational data (Section 10) suggests $Z(p) = O(1)$, far stronger than $O(p^{2/3})$.

Remark 12 (Content restriction). A stronger nonvanishing holds: if $p \geq 7$ divides the content of N_h (the gcd of all its coefficients), then $h \geq 2p$. Indeed, $N_h(-r) = (-1)^{r-1} b_{r-1} b_{h-r} \big((r-1)!\big)^3 \big((h-r)!\big)^3$ for $1 \leq r \leq h$ (by induction from the Apéry recurrence); choosing $r = 1, 2$ and using $p \nmid (h-1)!$, $(h-2)!$, 5 for $h \leq p$ recovers Lemma 8, while the range $p < h < 2p$ is excluded similarly via $r = h - p + 1$. Setting $\beta_n = (n!)^3 b_n$ and $F(z) = \sum_{n \geq 0} b_n z^n$, the endpoint factorization packages into the double generating function $\sum_{r,s \geq 0} \frac{N_{r+s+1}(-r-1)}{(r!)^3 (s!)^3} u^r v^s = F(-u)\, F(v)$, showing that the boundary array is a rank-one tensor.

Remark 13 (Adjacent resultant formula). The adjacent resultants satisfy the exact formula

$$\big|\mathrm{Res}(N_h, N_{h+1})\big| = \prod_{j=1}^{h-1} \big(j!^3\, b_j\big)^6. \tag{7}$$

This follows from the recursion $\mathrm{Res}(N_h, N_{h+1}) = N_h(-h)^6\ \mathrm{Res}(N_{h-1}, N_h)$ (since $N_{h+1} \equiv -(m+h)^6 N_{h-1} \pmod{N_h}$), combined with $N_h(-h) = (-1)^{h-1} N_h(-1) = (-1)^{h-1}((h-1)!)^3 b_{h-1}$ (the reflection identity and the first evaluation from Lemma 8). Verified for $h = 2, 3, 4, 5$; the cases $h = 2, 3, 4$ give $\mathrm{Res}(N_2, N_3) = -5^6$, $\mathrm{Res}(N_3, N_4) = -2^{18} \cdot 5^6 \cdot 73^6$, $\mathrm{Res}(N_4, N_5) = 2^{36} \cdot 3^{18} \cdot 5^{12} \cdot 17^{12} \cdot 73^6$.

A key consequence: $p \mid \mathrm{Res}(N_h, N_{h+1})$ if and only if $p \mid b_j$ for some $1 \leq j \leq h-1$ (when $p > h$, so factorials are coprime to p). In particular, N_2 and N_3 have disjoint root sets modulo p for all $p \geq 7$ (since $5^6 \not\equiv 0$).

Remark 14 (Sharpness of the bound). The exponent $2/3$ in Proposition 9 is optimal for any argument using only the degree bounds $\deg N_h = 3(h-1)$. The exact extremal problem $\max\big\{\sum r_h : r_h \leq 3(h-1),\ \sum h\, r_h \leq p\big\}$ has value $\frac{3}{2} p^{2/3} + O(p^{1/3})$, with leading constant $3/2$ achieved by filling gap sizes $h = 2, 3, \ldots, H$ to capacity and using the remaining space at $h = H+1$, where $H \sim p^{1/3}$. Moreover, degree bounds alone (even using all pairs of zeros, not just consecutive pairs) cannot yield $O(p^{1/2})$: an abstract polynomial model on $\sim p^{2/3}$ points can satisfy $\deg F_h \leq 3(h-1)$ for all relevant h. Improving the exponent requires genuinely arithmetic input about the family (N_h).

Remark 15 (Dodgson condensation). The Desnanot–Jacobi identity (Lewis Carroll's condensation) applied to the tridiagonal determinant gives the bilinear relation

$$N_{h+2}(m)\, N_h(m+1) - N_{h+1}(m)\, N_{h+1}(m+1) = -\prod_{j=2}^{h+1} (m+j)^6. \tag{8}$$

This is an identity in $\mathbb{Z}[m]$, verified for $h \leq 7$. A consequence: for $m \not\equiv -j \pmod{p}$ $(2 \leq j \leq h+1)$, the pairs $(m, h+2)$ and $(m+1, h)$ cannot both lie in the zero relation $\mathcal{Z} = \{(m,h) : N_h(m) \equiv 0\}$, since $N_{h+1}(m)\, N_{h+1}(m+1)$ would equal a nonzero product of sixth powers. After a meromorphic gauge $\tau_h(x) = N_h(x)/d_h(x)$ satisfying $d_{h+2}(x)\, d_h(x+1) = \prod_{j=2}^{h+1}(x+j)^6$, this becomes an exact SL_2-tiling: $\tau_{h+1}(x)\, \tau_{h+1}(x+1) - \tau_{h+2}(x)\, \tau_h(x+1) = 1$, equivalently an A_1 T-system (discrete Liouville equation). A zero of τ forces its transverse neighbors to be units (*singularity confinement*), but abstract SL_2-tilings can have zeros of density $1/2$ (e.g., $q_n = 0, 1, 0, -1, \ldots$ with $q_n^2 - q_{n-1} q_{n+1} = 1$), so the bilinear structure alone does not force a sublinear zero count. The arithmetic content—endpoint factorization, restart identity (Lemma 21), and polynomial degree bounds—is essential.

Remark 16 (Projective orbit reformulation). Let $B_n = n!^3\, b_n$ and D_n be the second solution of the renormalized recurrence $w_{n+1} = P(n)\, w_n - n^6\, w_{n-1}$ with $(D_0, D_1) = (0, 1)$. Their Casoratian is $B_n D_{n+1} - D_n B_{n+1} = (n!)^6$, nonzero for $n < p$. Define the *projective Apéry orbit* $\pi \colon \{0, \ldots, p-1\} \to \mathbb{P}^1(\mathbb{F}_p)$ by $\pi_n = [B_n : D_n]$. For $0 \leq m < m + h < p$, $N_h(m) \equiv 0 \pmod{p}$ if and only if $\pi_m = \pi_{m+h}$. Thus the zero-count $Z(p)$ equals the number of distinct collisions in the orbit π. The reflection symmetry $P(-1-n) = -P(n)$ implies $b_{p-1-n} \equiv b_n \pmod{p}$, making the orbit a palindrome: $\pi_n = \pi_{p-1-n}$. In particular, $Z(p)$ is even except at the non-ordinary primes $p \mid a_p(f)$ (at most two primes $p \le 10^5$).

Remark 17 (Galois structure). For each fixed h, the Galois group of the splitting field of N_h over $\mathbb{Q}$ is not the full symmetric group: the reflection $N_h(-m-h-1) = (-1)^{h-1}N_h(m)$ forces roots into pairs $\{r, -r-h-1\}$, and $\mathrm{Gal}(N_h) \cong C_2 \wr S_m$ (hyperoctahedral group) with $m = \lfloor 3(h-1)/2 \rfloor$ (confirmed exactly for $h \le 5$, $2 \le h \le 9$ by obstruction tests). By the Chebotarev density theorem, the average number of $\mathbb{F}_p$-roots of N_h is 1 for odd h and 2 for even h. This prediction is verified: for 262 primes in $[1000, 3000]$, the measured averages are $0.98, 2.03, 1.11, 2.07, 1.11, 1.88$ for $h = 3, \dots, 8$.

Proposition 18 (Hyperoctahedral Poisson law). *Assume* $\mathrm{Gal}(N_h) \cong C_2 \wr S_m$ *with* $m = \lfloor 3(h-1)/2 \rfloor$*. Let* $Y_h(p)$ *be the number of non-central reflection-pairs of roots of* N_h *that split completely over* $\mathbb{F}_p$*. Then for each fixed* $k \le m$*, the factorial moments in the prime-aspect satisfy*

$$\lim_{x\to\infty} \frac{1}{\pi(x)} \sum_{p \le x} (Y_h(p))_k = \left(\frac{1}{2}\right)^k,$$

where $(Y)_k = Y(Y-1)\cdots(Y-k+1)$*. Since these match the factorial moments of* $\mathrm{Poisson}(1/2)$ *for all* k*, the prime-aspect distribution of* $Y_h(p)$ *converges to* $\mathrm{Poisson}(1/2)$ *as* $h \to \infty$*.*

Proof. In $G = C_2 \wr S_m = \{(\sigma, \varepsilon) : \sigma \in S_m,\ \varepsilon \in \{\pm 1\}^m\}$, a *positive fixed point* of $g = (\sigma, \varepsilon)$ is an index i with $\sigma(i) = i$ and $\varepsilon_i = +1$. The Frobenius at p acts on the m reflection-pairs, and $Y_h(p)$ equals the number of positive fixed points. For any k distinct indices $i_1, \dots, i_k$, the fraction of $g \in G$ fixing all k positively is

$$\frac{2^{m-k}\,(m-k)!}{2^m\, m!} = \frac{1}{2^k\,(m)_k}.$$

Summing over all $(m)_k$ ordered k-tuples, $\mathbf{E}_{g\in G}[(Y(g))_k] = (1/2)^k$, which matches the factorial moments of $\mathrm{Poisson}(1/2)$. The Chebotarev density theorem then gives the prime-aspect convergence (Remark 20 supplies the effective error term under GRH). Since the factorial moments determine the Poisson distribution and they hold for all $k \le m$, $Y_h(p) \to \mathrm{Poisson}(1/2)$ as $m \to \infty$. □

Remark 19 (Prime-aspect vs. orbit-aspect). Proposition 18 is a *prime-aspect* statement: for fixed h and varying p, the root-pair count $Y_h(p)$ converges to $\mathrm{Poisson}(1/2)$. The zero-count $Z(p)$ aggregates contributions from all gaps $h = 1, \dots, p-1$ at a *fixed* prime p, so the orbit-aspect independence needed to deduce $Z(p)/2 \sim \mathrm{Poisson}(1/2)$ is not implied by Chebotarev alone. The gap between the two aspects parallels the gap between the Sato–Tate distribution (average over primes) and pointwise distribution (a single prime).

Remark 20 (GRH-effective error). For each fixed k, the k-th factorial moment does not require the full splitting field K_h (of degree $2^m m!$, double-exponential in h). Let $E_{h,k}$ be the fixed field of the stabilizer of an ordered signed k-tuple with distinct underlying reflection pairs; then $[E_{h,k} : \mathbb{Q}] = 2^k\,(m)_k = O_k(h^k)$. Under GRH for $\zeta_{E_{h,k}}$,

$$\frac{1}{\pi(x)} \sum_{\substack{p \le x \\ p \text{ good}}} (Y_h(p))_k = 2^{-k} + O_k\!\left(\frac{\log D_{h,k} + 2^k (m)_k \log x}{\sqrt{x}}\right),$$

where $D_{h,k} = |\operatorname{Disc}(E_{h,k})|$. The Apéry recurrence gives the coefficient-height bound $\log \|N_h\|_1 = O(h \log h)$, from which $\log D_{h,k} \ll_k h^{k+2} \log h$. Thus an $O(h^{-C})$ error holds in the polynomial range $x \ge h^{2k+4+2C+\varepsilon}$, for each fixed k.

At $x = 10^6$ with $h = 10$ ($m = 13$), the degree contributions alone for $k = 1, 2, 3$ are 0.25, 5.99, and 131.7 respectively: the observed Poisson accuracy is far better than present effective Chebotarev can certify.

5.2 Dual bounds: the restart identity and column estimate

The bound $Z(p) = O(p^{2/3})$ controls zeros along a *fixed row* (varying m for a given gap h). A dual bound controls zeros along a *fixed column* (varying h for a given starting point x_0).

Lemma 21 (Restart identity)**.** *Suppose $N_r(x_0) = 0$ and $N_{r+1}(x_0) \neq 0$ for some $x_0 \in \mathbb{F}_p$ and $r \geq 1$. Then for every $d \geq 0$,*

$$N_{r+d}(x_0) = N_{r+1}(x_0) \cdot N_d(x_0 + r). \tag{9}$$

In particular, if $N_{r+d}(x_0) = 0$ as well, then $N_d(x_0 + r) = 0$.

Proof. The sequence $Y_d = N_{r+d}(x_0)/N_{r+1}(x_0)$ has $Y_0 = 0$, $Y_1 = 1$, and satisfies $Y_{d+1} = P((x_0{+}r){+}d)\, Y_d - ((x_0{+}r){+}d)^6\, Y_{d-1}$, which is the defining recurrence for $N_d(x_0 + r)$. $\square$

Proposition 22 (Column bound)**.** *For every $x_0 \in \mathbb{F}_p$ and every interval I of H consecutive h-values with $H \leq p$,*

$$\#\{h \in I : N_h(x_0) \equiv 0 \pmod p\} \leq \tfrac{3^{4/3}}{2}\, H^{2/3} + O(H^{1/3} + 1). \tag{10}$$

Proof. Split I at the unique transition h where $x_0 + h \equiv 0 \pmod p$ (if it exists in I), creating at most two nonsingular pieces plus an $O(1)$ additive cost.

On a nonsingular piece, order the zero levels. For threshold $L \geq 2$: a gap of length $d \leq L$ between consecutive zeros at levels $r < r + d$ gives $N_d(x_0 + r) = 0$ by Lemma 21. For fixed d, there are at most $3(d{-}1)$ values of $x_0{+}r$ satisfying this (by Lemma 8 and $\deg N_d = 3(d{-}1)$). Gaps longer than L contribute at most $H/(L{+}1)$ zeros. Summing:

$$\text{zero count} \leq \frac{H}{L+1} + \sum_{d=2}^{L} 3(d{-}1) + O(1) = \frac{H}{L+1} + \tfrac{3}{2}L(L{-}1) + O(1).$$

Optimizing $L \asymp (H/3)^{1/3}$ gives the stated bound. $\square$

Computationally, the column count at mesoscopic scale $H = \lfloor\sqrt{p}\rfloor$ is far below the $O(H^{2/3})$ bound: for all primes $p \leq 100{,}003$ and all x_0, $\#\{h \leq H : N_h(x_0) \equiv 0\} \leq 2$. For $p \geq 10{,}007$, the maximum is ≤ 1.

Remark 23 (Rectangular incidence estimate). The row bound $|R_h| \leq 3(h{-}1)$ and the column bound (10) together give the rectangular incidence estimate

$$\#\{(x, h) : x \in \mathbb{F}_p,\ 2 \leq h \leq H,\ N_h(x) \equiv 0\} \leq \min\{H^2,\ pH^{2/3}\}.$$

The first bound sums row degrees; the second sums column estimates. They cross at $H \sim p^{3/4}$. Note that a uniform bound $R_p(H) \leq CH$ for all primes is *false*: by the Chebotarev density theorem, for each fixed H, infinitely many primes $p > H^2$ have every N_h ($2 \leq h \leq H$) splitting completely over $\mathbb{F}_p$, giving $R_p(H) = \frac{3}{2}H(H-1)$. The correct target is the *mesoscopic* estimate $R_p(\lfloor\sqrt{p}\rfloor) = O(\sqrt{p})$, where the split-prime obstruction does not apply because $H = \sqrt{p}$ grows with p. More generally, if $R_p(H) \ll H^\beta$ then $Z(p) \leq p/H + R_p(H)$ is optimized at $H \sim p^{1/(\beta+1)}$, giving $Z(p) \ll p^{\beta/(\beta+1)}$: the current $\beta = 2$ gives 2/3 and $\beta = 1$ yields 1/2. Reflection forces $R_p(H) \geq H/2$ from central roots, so $\beta = 1$ is the intrinsic limit of this first-moment method.

5.3 The continuant Bezout identity

Write $S_d(x) = \prod_{j=2}^{d}(x+j)$.

Proposition 24 (Bezout identity)**.** *For integers $2 \le d < e$,*

$$N_{e-1}(x{+}1)\, N_d(x) - N_{d-1}(x{+}1)\, N_e(x) = S_d(x)^6\, N_{e-d}(x+d). \tag{11}$$

In particular, after saturating the cut-edge divisor S_d,

$$(N_d, N_e) : S_d^{\infty} = (N_d,\ N_{e-d}(x+d)) : S_d^{\infty}.$$

Proof. The continuant addition formula (splitting the interval $[1, e{-}1]$ at position d) gives $N_e = N_d\, N_{e-d+1}(x{+}d{-}1) {-} (x{+}d)^6\, N_{d-1}\, N_{e-d}(x{+}d)$. The Dodgson identity at index $d{-}2$ gives $N_{d-1}(x{+}1)\, N_{d-1} - N_{d-2}(x{+}1)\, N_d = \prod_{j=2}^{d-1}(x{+}j)^6$. Combining these two identities yields (11). Verified computationally for all $2 \le d < e \le 9$. ☐

Corollary 25 (Separated-block criterion)**.** *Away from the cut-edge divisor $S_d(x) = 0$,*

$$N_d(\alpha) \equiv N_e(\alpha) \equiv 0 \pmod p \quad \Longleftrightarrow \quad N_d(\alpha) \equiv N_{e-d}(\alpha + d) \equiv 0 \pmod p.$$

In particular, $\deg \gcd_{\mathbb{F}_p}(N_d, N_e) \le 3(\min(d, e{-}d) - 1)$*, which improves the naive* $3(d{-}1)$ *when* $e - d < d$*.*

Remark 26 (Separated-block resultant)*.* Define $\mathcal{R}_{d,r} = \mathrm{Res}_x(N_d(x), N_r(x{+}d))$. The Bezout identity gives

$$|\operatorname{Res}(N_d, N_e)| = \prod_{j=1}^{d-1} \big((j!)^3 b_j\big)^6 \cdot |\mathcal{R}_{d,e-d}|.$$

For adjacent indices ($e = d+1$), $N_1 = 1$ gives $\mathcal{R}_{d,1} = 1$, recovering the adjacent resultant formula. For nonadjacent indices, $\mathcal{R}_{d,r}$ is genuinely new. The reflection identity gives $|\mathcal{R}_{d,r}| = |\mathcal{R}_{r,d}|$. The subtraction step does *not iterate*: after one application, $N_d(x)$ and $N_{e-d}(x + d)$ live on disjoint shifted intervals with opposite orientations, so no second nested division is possible. For $d = 2$, writing $P(y) = (2y{+}1)Q(y)$ with $Q(y) = 17y^2{+}17y{+}5$ and θ a root of Q, the norm decomposition $\mathcal{R}_{2,r} = 2^{3(r-1)} G_r(-\frac{1}{2}) \cdot 17^{3(r-1)}\, \mathrm{N}_{\mathbb{Q}(\theta)/\mathbb{Q}}(G_r(\theta))$ (where $G_r = N_r(x+1)$) gives $\mathcal{R}_{2,2} = 2^5 \cdot 5^4 \cdot 11^2 \cdot 17^3 \cdot 71$. At $p = 71$: $N_2(-5/2) \equiv N_4(-5/2) \equiv 0 \pmod{71}$ with $-5/2 \notin \{-1, -2, -3, -4\}$, a nonboundary common root inside the mesoscopic range ($4^3 < 71$). This shows the boundary-only gcd statement is *false*.

Remark 27 (Collision energy and the $H^{1/2}$ barrier)*.* The column exponent $2/3$ is the natural barrier for the consecutive-gap argument: even if each gap of length d has only d (rather than $3(d{-}1)$) admissible starts, the sum $\sum_{d \le L} d \asymp L^2$ still gives $H/L + L^2$, optimized at $H^{2/3}$. To achieve $H^{1/2}$, one needs the aggregate collision energy $\mathcal{E}_p(H) = \sum_{d+r \le H} \#\{x : N_d(x) \equiv N_r(x{+}d) \equiv 0\} \ll H^{3/2}$, which is a sparse-exception theorem for the separated-block resultant family—beyond the scope of universal continuant identities. (The A_1 T-system $\tau_{h+1}(x)\tau_{h+1}(x+1) - \tau_{h+2}(x)\tau_h(x+1) = 1$ admits tame solutions with zero density 1/2, e.g. $\tau_{2k} = 0$, $\tau_{2k+1} = (-1)^k$, so singularity confinement alone cannot force sparsity.) In the transfer-matrix formulation (writing $M(n) = \left(\begin{smallmatrix} P(n) & -n^6 \\ 1 & 0 \end{smallmatrix}\right)$, so that $N_h(x) = 0$ iff a fixed matrix coefficient of the interval product $M(x{+}h{-}1) \cdots M(x{+}1)$ vanishes), the conjectured $\mathcal{E}_p(H) \ll H^{3/2}$ is analogous to a Rudnev-type point–plane incidence bound in $\mathrm{PGL}_2(\mathbb{F}_p)$, subject to a nonconcentration hypothesis for the deterministic Apéry cocycle. Computation for selected primes:

p	71	503	2003	5003	7919	20011	50021	$10^5{+}3$
$H = \lfloor\sqrt{p}\rfloor$	8	22	44	70	88	141	223	316
$R_p(H)/H$	1.75	1.59	1.36	1.50	1.75	1.63	1.56	1.46
$\mathcal{E}_p(H)$	1	2	2	2	10	0	2	0
$\mathcal{E}_p(H)/H^{3/2}$	.044	.019	.007	.003	.012	0	.001	0

The ratio $R_p(H)/H$ clusters around 3/2 (the Chebotarev prediction from Remark 17), and $\mathcal{E}_p(H)/H^{3/2} \to 0$, consistent with $\mathcal{E}_p(H) \ll H^{3/2}$. The key arithmetic object is the shifted resultant $S_{d,r} = \mathrm{Res}_x(N_d(x), N_r(x{+}d)) \in \mathbb{Z}$: for $p \nmid S_{d,r}$, the gap polynomials N_d and $N_r(\cdot{+}d)$ share no root over $\mathbb{F}_p$, so that gap pair contributes zero to $\mathcal{E}_p$. A Hadamard bound gives $\log|S_{d,r}| \ll (d{+}r)^2 \log(d{+}r)$, so the total number of prime divisors across all pairs (d, r) with $d{+}r \le H$ is $O(H^4 \log H)$. Averaging over the $\sim X/\log X$ primes in $(X, 2X]$ gives $\mathcal{E}_p(H) \ll H^{1/2}$ for all but $o(X/\log X)$ primes, provided $H \le X^{2/7-\varepsilon}$—rigorous but far from the global target $H = p - 1$.

A sharper route uses the bipartite incidence graph $\Gamma_p(H)$: left vertices are gap lengths d, right vertices are gap lengths r, with an edge when N_d and $N_r(\cdot + d)$ share a root over $\mathbb{F}_p$. Each edge certifies a triple collision, so $\mathcal{E}_p(H) = e(\Gamma_p(H))$. A uniform fiber bound $\deg_L(d) \le C$ (the number of r-partners for each d) and a uniform codegree bound (at most C common right neighbors for any two left vertices) would place $\Gamma_p(H)$ in the regime of the Kővári–Sós–Turán theorem, giving $\mathcal{E}_p(H) \ll H^{3/2}$—the collision-energy scale needed for the mesoscopic target $R_p(\sqrt{p}) = O(\sqrt{p})$. The Dodgson identity (Remark 15) restricts local configurations in Γ_p, but by itself controls $O(H)$ pairs out of $O(H^2)$; upgrading it to a global codegree bound is the key missing step.

5.4 Computational evidence

Proposition 28. *For all* 78,496 *primes* $5 \le p \le 10^6$*:*

1. *$Z(p) \in \{0, 2, 4, 6, 8, 10, 12\}$ for all but two primes; the two exceptions ($p = 11$ and $p = 3137$) have $Z(p) = 1$ (non-ordinary primes, where $p \mid a_p(f)$).* $\max Z(p) = 12$ *at* $p = 159{,}977$.

2. *The exact histogram is:*

$Z(p)$	*0*	*1*	*2*	*4*	*6*	*8*	*10*	*12*
count	*47632*	*2*	*23730*	*6045*	*951*	*123*	*12*	*1*
fraction	*.607*	*.000*	*.302*	*.077*	*.012*	*.002*	*.000*	*.000*
Poi$(\frac{1}{2})$	*.607*	—	*.303*	*.076*	*.013*	*.002*	*.000*	*.000*

The mean $Z(p) = 1.000$, stable across all ranges ($\chi^2 = 3.42$ on 4 d.f. against Poisson(1/2) *for the pair count). The empirical factorial moments of $K(p) = Z(p)/2$ over the* 78,494 *ordinary primes are:*

r	*1*	*2*	*3*	*4*	*5*	*6*
$\frac{1}{\pi_{\mathrm{ord}}}\sum (K)_r$	.4998	.2490	.1210	.0605	.0275	.0092
2^{-r}	.5000	.2500	.1250	.0625	.0312	.0156

The first four moments match Poisson(1/2) to better than 3%*; the higher-moment deviation is consistent with sampling variance ($K \ge 5$ has only* 13 *primes in this range).*

3. *The symmetry $b_j \equiv b_{p-1-j} \pmod{p}$ holds for all tested primes: zeros come in palindromic pairs $\{j, p{-}1{-}j\}$, so $Z(p)$ is even (except at non-ordinary primes). The pair count $Z(p)/2$ fits* Poisson(1/2) *to within sampling noise (table above). Proposition 18 proves* Poisson(1/2) *rigorously in the prime-aspect; the orbit-aspect fit (Remark 19) remains empirical.*

Remark 29 (Mesoscopic root count). At the scale $H = \lfloor\sqrt{p}\rfloor$, the total root count $R_p(H) = \sum_{h=2}^{H} r_p(h)$ satisfies $R_p(H)/H \approx 3/2$ for all tested primes, consistent with the Chebotarev prediction from Remark 17:

p	$H = \lfloor\sqrt{p}\rfloor$	$R_p(H)$	$R_p(H)/H$
503	22	35	1.59
2003	44	60	1.36
5003	70	105	1.50
10007	100	154	1.54
20011	141	230	1.63
50021	223	347	1.56
100003	316	460	1.46

The ratio $R_p(H)/H$ stabilizes near 3/2 (range 1.32–1.75 for $p > 500$), supporting the mesoscopic conjecture in Remark 23.

5.5 Hypothesis Z

Hypothesis 30 (Hypothesis Z)**.** *There exists an absolute constant C such that $Z(p) \leq C$ for all primes p.*

Remark 31. The analogous problem for the $\zeta(2)$ Apéry numbers $A_n = \sum_k \binom{n}{k}^2\binom{n+k}{k}$ has a fundamentally different structure. Stienstra–Beukers [20] showed that A_p is related to a *CM* modular form, yielding $A_{(p-1)/2} \equiv 0 \pmod{p}$ for all $p \equiv 3 \pmod{4}$ — a *positive density* of primes with guaranteed zeros. In contrast, the $\zeta(3)$ Apéry numbers are governed by the non-CM newform $f(z) = \eta(2z)^4\eta(4z)^4 \in S_4(\Gamma_0(8))$ (LMFDB label 8.4.a.a), for which no such systematic zero-forcing occurs. This is a structural reason why Hypothesis $\bar{Z}$ should hold for the $\zeta(3)$ sequence, even though the pointwise bound $Z(p) \leq C$ (Hypothesis Z) likely fails.

Remark 32. Even the single-row slice $Z^*(p) := \mathbf{1}[p \mid b_{(p-1)/2}]$ is connected to the non-ordinary primes of f via the Beukers congruence $b_{(p-1)/2} \equiv a_p(f) \pmod{p}$ [5, 1]. The density-zero conjecture for non-ordinary primes of non-CM weight-4 forms is itself open [14]. Thus Hypothesis Z contains a known open problem as a special case. The heuristic density of non-ordinary primes is $\sim C_f/p$ (the Sato–Tate distribution on the normalized coefficients $a_p/p^{3/2}$ admits $\sim 4p^{1/2}$ multiples of p in $[-2p^{3/2}, 2p^{3/2}]$, each contributing mass $\sim p^{-3/2}$). The expected count up to x is thus $C_f \log\log x$ with $C_f \approx 1$. Finding 2 non-ordinary primes ($p = 11, 3137$) below 10^6 is consistent with $\log\log(10^6) \approx 2.9$. This is not a standard Lang–Trotter problem: the target set $\{a : p \mid a\}$ varies with p, and no analogue of Elkies' theorem (infinitely many supersingular primes) is known for weight ≥ 4.

Remark 33 (Evidence against Hypothesis Z). The $2 \cdot \mathrm{Poisson}(1/2)$ distribution (Remark 7 below) has unbounded support: extreme-value theory predicts $\max_{p \leq N} Z(p) \sim 2\log\log N$. The current record $Z(p) = 12$ at $p = 159{,}977$ (with record progression $2, 4, 6, 8, 10, 12$) is consistent with this prediction. We therefore expect Hypothesis Z to fail: the first prime with $Z(p) = 14$ should appear near $p \approx 2 \times 10^7$ (the Gumbel prediction for the Poisson(1/2) pair maximum). A computation to $p = 5 \times 10^7$ would either refute Hypothesis Z or provide strong evidence for a structural cutoff.

Hypothesis 34 (Hypothesis $\bar{Z}$)**.** $\displaystyle\sum_{p \leq x} Z(p) \ll \pi(x)$*, equivalently, the average of $Z(p)$ over primes $p \leq x$ is bounded.*

Remark 35. Hypothesis $\bar{Z}$ is strictly weaker than Hypothesis Z and consistent with $Z(p) \sim 2 \cdot \mathrm{Poisson}(1/2)$ (which gives $\mathbb{E}[Z(p)] = 1$, independent of p). It suffices for the conditional bound $\log G_n = O(\sqrt{n})$ (Theorem 55).

We state the full distributional prediction of the Poisson model (cf. Proposition 18):

Hypothesis 36 (Poisson conjecture)**.** *For every good prime $p \geq 5$, put*

$$\nu_p := \mathbf{1}_{\{p|a_p(f)\}}, \qquad K_*(p) := \frac{Z(p) - \nu_p}{2},$$

where f is the weight-4 newform associated to the Apéry differential operator. Thus $K_*(p)$ counts noncentral reflected pairs of zeros of $(b_j \bmod p)_{0 \leq j < p}$. For every fixed $r \geq 1$,

$$\frac{1}{\pi(X)} \sum_{p \leq X} \binom{K_*(p)}{r} \longrightarrow \frac{1}{2^r\, r!}.$$

On ordinary primes ($\nu_p = 0$), this reduces to $K(p) = Z(p)/2 \to \mathrm{Poisson}(1/2)$, consistent with the data in Proposition 28.

Remark 37. The Poisson conjecture implies Hypothesis $\bar{Z}$. Taking $r = 1$ gives $\sum_{p \leq X} K_*(p) = (\frac{1}{2} + o(1))\,\pi(X)$. Since $Z(p) = 2K_*(p) + \nu_p$ and $0 \leq \nu_p \leq 1$,

$$\sum_{p \leq X} Z(p) = 2 \sum_{p \leq X} K_*(p) + \sum_{p \leq X} \nu_p \leq (2 + o(1))\,\pi(X),$$

which is Hypothesis $\bar{Z}$. A proof would require the fixed-order joint independence of reflection-pair zero events—the correlation theorem underlying the factorial-moment computation in Proposition 18—for the deterministic Apéry orbit, not just the hyperoctahedral random model.

Remark 38 (Computational evidence for $\bar{Z}$). Over dyadic blocks of primes, $(\sum_{P<p\leq 2P} Z(p))/|\{P < p \leq 2P\}|$ ranges from 0.86 to 1.07 for $P = 2^k$, $3 \leq k \leq 12$ (666 primes up to 4999), with overall average $\sum Z(p)/\pi = 0.91$. The Poisson prediction is 1.

6 The denominator connection lemma

Lemma 39. *Let $p \geq 7$ be prime with $n/2 < p \leq n$, and set $r = n - p$. Then $v_p(G_n) = 0$ if and only if $p \nmid b_r$.*

Proof. Since $p > n/2$, the only multiple of p in $\{1, \dots, n\}$ is p itself, so $v_p(d_n) = 3$.

Step 1. $v_p(d_n b_n) = 3 + v_p(b_n) \geq 3$, so $v_p(G_n) = 0$ requires $v_p(d_n a_n) = 0$, i.e., $v_p(a_n) = -3$.

Step 2 (variation of parameters). From Lemma 1, the Casorati identity telescopes to

$$\frac{a_n}{b_n} = \sum_{m=1}^{n} \frac{6}{m^3\, b_m\, b_{m-1}}. \tag{12}$$

For $p > n/2$, the only $m \in \{1, \dots, n\}$ with $p \mid m$ is $m = p$.

Step 3. By Theorem 4, $b_p \equiv b_0\, b_1 = 5 \pmod{p}$ (since p has base-p digits $(0, 1)$), and by the symmetry $b_j \equiv b_{p-1-j}$, $b_{p-1} \equiv b_0 = 1 \pmod{p}$. Since $p \geq 7$, both are nonzero mod p.

The $m = p$ term in (12) has $v_p = -3$; all other terms have $v_p \geq 0$. Therefore

$$p^3 \cdot \frac{a_n}{b_n} \equiv \frac{6}{b_p\, b_{p-1}} \equiv \frac{6}{5} \pmod{p}.$$

Using $b_n \equiv 5\,b_r \pmod p$ (Theorem 4, since $n = 1\cdot p + r$):

$$p^3\,a_n \equiv \tfrac{6}{5}\cdot 5\,b_r = 6\,b_r \pmod p.$$

Thus $v_p(a_n) = -3$ iff $p \nmid b_r$. $\square$

Verification. For all $n \le 300$ and primes $p \in (n/2, n]$ with $p \ge 7$: $v_p(G_n) = 0 \iff p \nmid b_{n-p}$ confirmed in 2101/2101 cases (100%).

Corollary 40 (Lucas reduction)**.** *For $p \ge 7$ and $p \in (n/2, n]$: $v_p(G_n) \ge 1$ iff $p \mid b_{n-p}$. By the Lucas congruence $b_n \equiv b_1\,b_{n-p} = 5\,b_{n-p} \pmod p$, this is equivalent (for $p \ne 5$) to $p \mid b_n$. Thus $B(n) := \#\{p \in (n/2, n], p \text{ prime} : v_p(G_n) \ge 1\}$ equals $\omega_{(n/2,n]}(b_n) + O(1)$, the number of* distinct prime factors of b_n lying between $n/2$ and n*, up to an absolute constant.*

The full conjecture $G_n = e^{o(n)}$ for all n is therefore equivalent to the *top-half radical estimate*:

$$\log \operatorname{rad}_{(n/2,n]}(b_n) = o(n). \tag{13}$$

Since $\log b_n \sim n\log(17 + 12\sqrt{2}) \approx 3.52\,n$, the height bound gives $\omega_{(n/2,n]}(b_n) \le \log b_n/\log(n/2) = O(n/\log n)$, precisely one $o(1)$ factor short of the target. Computation for $n \le 200{,}000$ gives $B(n) \le 3$ and $B(n) = 0$ for 93.7% of values (see Remark 41 for the Poisson analysis).

Remark 41 (Random-model prediction)*.* Under a random-integer model, $\Pr(p \mid b_n) \approx 1/p$ for each prime p, so by Mertens' theorem

$$\mathbb{E}[B(n)] = \sum_{n/2<p\le n} \frac{1}{p} = \log\Big(\frac{\log n}{\log(n/2)}\Big) + O(1/\log n) \approx \frac{\log 2}{\log n} \;\to\; 0.$$

Computation for $n \le 200{,}000$ confirms: $\mathbb{E}[B] \approx 0.065$, $\max B(n) = 3$, and the histogram matches Poisson($\log 2/\log n$) to high accuracy: $B(n) = 0$ for 93.7%, $B(n) = 1$ for 6.0%, $B(n) = 2$ for 0.20%, $B(n) = 3$ for 0.006% (the Poisson predictions with $\lambda = 0.065$ are 93.7%, 6.1%, 0.20%, 0.004%). The dispersion ratio $\operatorname{Var}(B)/\mathbb{E}[B] = 1.003$ is the exact Poisson signature. By CRT-based approximate independence across primes, $B(n)$ should follow Poisson($\log 2/\log n$) with $\lambda \to 0$. Extreme-value theory then predicts $\max_{n\le N} B(n) = O(\log N/\log\log N)$—trivially $o(n/\log n)$—so the random model *strongly* predicts the full conjecture. The obstruction is proving that b_n behaves "randomly" with respect to its prime factors in $(n/2, n]$, despite its highly structured factorization (supercongruences, Lucas decomposition, palindromic symmetry).

7 The unconditional density-one theorem

Theorem 42. *For a set of positive integers n of natural density 1, $G_n = e^{o(n)}$.*

Proof. Fix N large and consider $n \in (N, 2N]$. Write $B(n) = \#\{\text{primes } p > \sqrt{n} \text{ with } v_p(G_n) \ge 1\}$. By Proposition 3, the small-prime contribution is $O(\sqrt{n})$ for every n. Each such prime contributes $v_p(G_n)\log p \le 3\log n$ (since $v_p(d_n) = 3$ for $p > \sqrt{n}$), so $\log G_n \le O(\sqrt{n}) + 3B(n)\log n$. It suffices to show $B(n) = o(n/\log n)$ for all but $o(N)$ values of $n \in (N, 2N]$.

For $p > \sqrt{n}$, write $n = qp + r$ with $1 \le q \le \sqrt{2N}$ and $0 \le r < p$. By Lemma 47(iii), $v_p(G_n) \ge 1$ requires $D_q b_r \equiv 0 \pmod p$. By Lemma 48, leading-digit bad primes ($p \mid b_q$, $p \nmid b_r$) have $v_p(G_n) = 0$ and do not contribute to $B(n)$. The remaining contributions come from *lower-digit bad primes* ($r \in \mathcal{Z}_p$) and *companion-block primes* ($D_q \equiv 0 \pmod p$, $p \nmid b_r$).

Lower-digit incidence. For fixed $p \in (\sqrt{N}, 2N]$, an interval of N consecutive integers contains at most $N \cdot Z(p)/p + Z(p)$ values of n whose residue $r = n \bmod p$ lies in $\mathcal{Z}_p$. Set $\rho_p = Z(p)/p$. The total lower-digit incidence is

$$R_N \leq \sum_{\sqrt{N}<p\leq 2N} \big(N\rho_p + Z(p)\big).$$

By Proposition 9, $\rho_p \to 0$. Since $\pi(2N) = O(N/\log N)$:

$$\sum_p \rho_p = o(\pi(2N)) = o(N/\log N), \qquad \sum_p Z(p) = \sum_p p\rho_p \leq 2N \sum_p \rho_p = o(N^2/\log N).$$

Hence $R_N = o(N^2/\log N)$.

Companion-block incidence. By Proposition 51, the companion-block contribution to $\log G_n$ is $O(n^{2/3}) = o(n)$ at every index. Hence companion-block primes do not affect the density-one statement, and we set $P_N = 0$ in the counting argument.

Conclusion. Write $\sum_{n\in(N,2N]} B(n) \leq R_N + P_N = o(N^2/\log N)$. For any fixed $\delta > 0$, Markov's inequality gives

$$\#\big\{n \in (N, 2N] : B(n) > \delta n/\log n\big\} \leq \frac{o(N^2/\log N)}{\delta N/\log N} = o(N).$$

A dyadic decomposition shows the exceptional set has density zero, so $B(n) = o(n/\log n)$ for density 1. For all such n, $\log G_n = O(\sqrt{n}) + 3B(n)\log n = O(\sqrt{n}) + o(n) = o(n)$. □

Corollary 43 (Quantitative exceptional set)**.** *For every $\varepsilon > 0$,*

$$\#\{n \leq N : \log G_n > \varepsilon\, n\} = O_\varepsilon(N^{2/3}).$$

Proof. We make the density-one argument quantitative using the explicit bound $Z(p) = O(p^{2/3})$ from Proposition 9. For $n \in (N, 2N]$, $\log G_n \leq O(\sqrt{n}) + 3B(n)\log n$, so $\log G_n > \varepsilon n$ requires $B(n) > \varepsilon N/(7\log N)$ for N large.

Partial summation with $Z(p) \leq Cp^{2/3}$ gives

$$\sum_{\sqrt{N}<p\leq 2N} \frac{Z(p)}{p} \leq C \sum_{p\leq 2N} p^{-1/3} = \tfrac{3}{2}\, C\, \frac{(2N)^{2/3}}{\log(2N)} + O(N^{1/3}) = O\left(\frac{N^{2/3}}{\log N}\right).$$

The lower-digit incidence satisfies $R_N = N\sum Z(p)/p + \sum Z(p) = O(N^{5/3}/\log N)$ (the second sum, $\sum Z(p) \leq C\sum p^{2/3}$, contributes the same order). The companion-block incidence $P_N = O(N^{3/2})$ is dominated by R_N. Hence $\sum_{n\in(N,2N]} B(n) = O(N^{5/3}/\log N)$.

By Markov's inequality,

$$\#\big\{n \in (N, 2N] : B(n) > \varepsilon N/(7\log N)\big\} \leq \frac{O(N^{5/3}/\log N)}{\varepsilon N/(7\log N)} = O(N^{2/3}/\varepsilon).$$

A dyadic decomposition over intervals $(N/2^{j+1}, N/2^j]$ gives the global bound, since $\sum_{j\geq 0}(N/2^j)^{2/3} = O(N^{2/3})$. □

In particular, the exceptional set has finite harmonic weight:

Corollary 44 (Finite harmonic weight)**.** *For every $\varepsilon > 0$,*

$$\sum_{\substack{n\ge 1\\ \log G_n > \varepsilon n}} \frac{1}{n} < \infty.$$

Proof. The contribution from $(2^k, 2^{k+1}]$ is $O(2^{-k/3})$ by Corollary 43, and the geometric series converges. $\square$

Remark 45. The exponent $2/3$ in the exceptional set is inherited from the lower-digit incidence R_N, which uses $Z(p) = O(p^{2/3})$. More generally, if $Z_p(I) \le C|I|^\alpha$ uniformly for all primes and sub-intervals (as Corollary 10 provides with $\alpha = 2/3$), then $R_N = O(N^{1+\alpha}/\log N)$ and $L_N = O(N^{(4-\alpha)/(3-\alpha)}/\log N)$. The exceptional set is therefore $O(\max(N^\alpha,\ N^{1/(3-\alpha)})/\varepsilon)$, with the crossover at $\alpha = (3-\sqrt{5})/2 \approx 0.382$. Under Hypothesis $\bar{Z}$, the dyadic bound $D(P,Q) \le \sum Z(p) = O(P/\log P)$ is equivalent to $\alpha = 0$ in this formula, giving the exceptional set $O(N^{1/3}/\varepsilon)$ (Corollary 57). Theorem 55 shows the matching density-one bound $\log G_n = O(\sqrt{n})$, where $O(\sqrt{n})$ is now the small-prime floor, not the leading-digit contribution.

8 Codegree amplification

We begin with a structural rigidity lemma: consecutive integers cannot share a top-half bad prime.

Lemma 46 (Consecutive coprimality)**.** *Let $p > 5$ be prime and $n \ge p$. If $p \mid b_{n \bmod p}$ and $p \mid b_{(n+1) \bmod p}$, then $p \le n/2$. Equivalently, no prime $p > n/2$ with $p > 5$ can be lower-digit bad for both n and $n+1$.*

Proof. Since $p > n/2$, both n and $n+1$ have base-p representations $n = p+r$, $n+1 = p+(r+1)$ with $0 \le r < r+1 < p$ (provided $r < p-1$; the case $r = p-1$ gives $n+1 = 2p$ and $b_{n+1} \equiv b_2 b_0 = 73 \not\equiv 0$ for $p > 73$, a finite check).

Suppose $p \mid b_r$ and $p \mid b_{r+1}$ with $r, r+1 < p$. The Apéry recurrence gives

$$(k+1)^3 b_{k+1} = P(k) b_k - k^3 b_{k-1},$$

and since $k+1 < p$ implies $(k+1)^3$ is invertible mod p, the conditions $p \mid b_k$ and $p \mid b_{k+1}$ force $p \mid b_{k-1}$. Iterating downward from $k = r$ gives $p \mid b_j$ for all $0 \le j \le r+1$, contradicting $b_0 = 1$. $\square$

A prime that divides b_n only through a leading base-p digit contributes nothing to G_n.

Lemma 47 (Block system for $p^3 a_n \bmod p$)**.** *Let $p \ge 7$, $p \le n < p^2$, and write $n = qp + r$ with $1 \le q < p$ and $0 \le r < p$. Define $\Delta_{q,r} \equiv p^3 a_{qp+r} \pmod{p}$ and $D_q = \Delta_{q,0}$. Then:*

(i) $\Delta_{q,r} \equiv D_q b_r \pmod{p}$ *for every $0 \le r < p$;*

(ii) $D_q b_{q-1} - D_{q-1} b_q \equiv 6/q^3 \pmod{p}$*;*

(iii) $p \mid G_n$ *if and only if* $D_q b_r \equiv 0 \pmod{p}$.

Proof. For $p > \sqrt{n}$, $v_p(\mathrm{lcm}(1,\ldots,n)) = 1$, so $v_p(d_n) = 3$. Since $d_n a_n \in \mathbf{Z}$, $p^3 a_n \in \mathbf{Z}_p$, and $v_p(d_n a_n) = v_p(p^3 a_n)$. Meanwhile $v_p(d_n b_n) = 3 + v_p(b_n) \ge 3$, so $p \mid G_n \Leftrightarrow p \mid p^3 a_n$.

(i) Multiply the Apéry recurrence at index $qp+r$ by p^3 and reduce modulo p. Since $qp + r \equiv r \pmod{p}$, the sequence $r \mapsto \Delta_{q,r}$ satisfies the same recurrence $(r+1)^3 \Delta_{q,r+1} = P(r)\Delta_{q,r} - r^3 \Delta_{q,r-1}$

as $r \mapsto D_q b_r$, with matching initial values $\Delta_{q,0} = D_q$ and $\Delta_{q,1} = 5D_q = D_q b_1$. Since $(r+1)^3$ is invertible mod p for $0 \le r \le p-2$, uniqueness gives the identity.

(ii) Multiply the Wronskian $a_n b_{n-1} - a_{n-1} b_n = 6/n^3$ at $n = qp$ by p^3 and reduce mod p, using $b_{qp} \equiv b_q$ and $b_{qp-1} \equiv b_{q-1} b_{p-1} \equiv b_{q-1}$ (Lucas), and $p^3 a_{qp-1} \equiv D_{q-1} b_{p-1} \equiv D_{q-1}$ (part (i)).

(iii) Immediate from (i) and the observation above. □

Lemma 48 (Leading-digit vanishing)**.** *Let $p \ge 7$, $p \le n < p^2$, $n = qp + r$ with $q \ge 1$. If $p \mid b_q$ and $p \nmid b_r$, then $v_p(G_n) = 0$.*

Proof. Since consecutive Apéry numbers below the p-th cannot both vanish mod p (the recurrence propagates to $b_0 = 1$), $p \nmid b_{q-1}$. Part (ii) of Lemma 47 gives $D_q b_{q-1} \equiv 6/q^3 \not\equiv 0 \pmod p$, so $D_q \not\equiv 0$. By part (iii), $p \mid G_n$ iff $D_q b_r \equiv 0$, but $D_q \not\equiv 0$ and $p \nmid b_r$, so $D_q b_r \not\equiv 0$ and $v_p(G_n) = 0$. □

Remark 49 (Block constants are Apéry numbers). The block constant D_q of Lemma 47 equals $a_q \bmod p$, where a_q is the Apéry rational companion. Indeed, a_q satisfies the same Wronskian recurrence $a_q b_{q-1} - a_{q-1} b_q = 6/q^3$ with $a_0 = 0$, $a_1 = 6$, so $D_q = a_q$ by uniqueness. Consequently, $D_q \equiv 0 \pmod p$ iff p divides the numerator of a_q (in lowest terms) and p does not divide its denominator.

Since $|a_q| \le C(1+\sqrt{2})^{4q}$ and $\mathrm{denom}(a_q) \mid \mathrm{lcm}(1,\dots,q)^3$, the numerator of a_q has $O(q)$ prime divisors. In particular, for fixed q, at most $O(q)$ primes p can be companion-block bad at quotient q. Computation through $n = 2000$ confirms that the companion-block count $\#\{p > \sqrt{n} : p \mid \mathrm{num}(a_{\lfloor n/p \rfloor})\}$ is at most 3 for every n in this range.

Proposition 50 (Reflection of the companion solution)**.** *Let $p \ge 5$ be prime. Every p-integral solution (u_n) of the Apéry recurrence* (1) *on $\{0,\dots,p-1\}$ is palindromic modulo p:*

$$u_{p-1-j} \equiv u_j \pmod p \qquad (0 \le j \le p-1).$$

In particular, for the companion solution $a_0 = 0$, $a_1 = 6$,

$$a_{p-1-j} \equiv a_j \pmod p, \qquad a_{p-1} \equiv 0 \pmod p.$$

Proof. Let R denote the reversal operator, $(Ru)_j = u_{p-1-j}$. The coefficient polynomial $P(n) = 34n^3 + 51n^2 + 27n + 5$ satisfies $P(-1-x) = -P(x)$, so substituting $n \mapsto p-2-n$ in the recurrence shows that R maps p-integral solutions to p-integral solutions modulo p.

The integer Apéry solution (b_n) satisfies $b_{p-1} \equiv 1$ and $b_{p-2} \equiv 5 \pmod p$: in the binomial sum $b_n = \sum_k \binom{n}{k}^2 \binom{n+k}{k}^2$, only the $k = 0$ term survives modulo p for $n = p-1$, while for $n = p-2$ only $k = 0, 1$ survive, contributing $1 + 4 = 5$. Since Rb and b have the same initial values $(b_0, b_1) = (1, 5)$, uniqueness gives $Rb = b$.

For two solutions x, y, the discrete Wronskian $W_n(x,y) = x_n y_{n-1} - x_{n-1} y_n$ satisfies $(n+1)^3 W_{n+1} = n^3 W_n$, so $W_n(x,y) = C(x,y)/n^3$. Using $Rb = b$:

$$W_n(Ru, b) = -W_{p-n}(u, b) = W_n(u, b).$$

Hence $W_n(Ru - u, b) = 0$, so $Ru - u = cb$ for some $c \in \mathbf{F}_p$. Applying R: on one hand $R(Ru-u) = -(Ru-u)$, on the other $R(cb) = cb$. Since p is odd, $c = 0$ and $Ru = u$. □

Proposition 51 (Companion-block height bound)**.** *Assume $D_q \equiv a_q \pmod p$. For every $n \ge 1$, the total logarithmic weight of companion-block primes is at most $O(n^{2/3})$:*

$$\sum_{\substack{p > \sqrt{n} \\ p \mid \mathrm{num}(a_{\lfloor n/p \rfloor})}} \log p \;\le\; C\, n^{2/3}$$

for an absolute constant C. In particular, companion-block primes contribute $o(n)$ to $\log G_n$ at every index.

Proof. Write $a_q = A_q/C_q$ in lowest terms. From the telescoping identity $a_q = b_q \sum_{k=1}^{q} 6/(k^3 b_k b_{k-1})$ and $b_k \geq 1$, we obtain $|A_q| \leq |a_q| \cdot C_q \leq C_0(1+\sqrt{2})^{4q}$, hence each numerator A_q has at most $O(q)$ prime divisors.

For a fixed n, set $Q = \lfloor n^{1/3} \rfloor$. A companion-block prime $p > \sqrt{n}$ has quotient $q = \lfloor n/p \rfloor < \sqrt{n}$.

Small quotients ($q \leq Q$): for each q, the primes dividing A_q contribute $\sum \log p \leq \log |A_q| = O(q)$. Summing over $q \leq Q$: $\sum_{q=1}^{Q} O(q) = O(Q^2) = O(n^{2/3})$.

Large quotients ($q > Q$): each such prime satisfies $p < n/Q = n^{2/3}$. Since each prime determines its quotient uniquely, the total log-weight from this range is bounded by the Chebyshev sum $\sum_{p<n^{2/3}} \log p = \theta(n^{2/3}) = O(n^{2/3})$. □

The first-moment bound in Corollary 43 can be sharpened to a polylogarithmic exceptional set by exploiting the gap polynomials a second time, now as a *codegree* bound between nearby integers.

Lemma 52 (Common bad-prime codegree)**.** *Let $m, n \in (N, 2N]$ with $h = n - m \geq 1$, and let $P_0 \geq 1$. The number of primes $p > P_0$ that are simultaneously lower-digit bad for m and for n (i.e., $p \mid b_{m \bmod p}$ and $p \mid b_{n \bmod p}$) is at most $O(h \log N / \log P_0)$. In particular, if $P_0 \geq N^{\delta}$ for any fixed $\delta > 0$, the codegree is $O(h)$.*

Proof. Since $p > N/2 \geq h$, the residues $m \bmod p$ and $n \bmod p$ lie in $\{0, \ldots, p-1\}$ and differ by exactly h. By the gap implication (Lemma 7), if $p \mid b_{m \bmod p}$ and $p \mid b_{n \bmod p}$, then $p \mid N_h(m \bmod p)$. The polynomial periodicity of N_h gives $N_h(m \bmod p) \equiv N_h(m) \pmod{p}$, so $p \mid N_h(m)$.

The gap polynomial has degree $3(h-1)$ in m (the tridiagonal determinant representation gives exactly this), so for $m \in (N, 2N]$, $|N_h(m)| \leq (CN^3)^h$. Since $N_h(m) > 0$ (the associated tridiagonal matrix is positive definite at positive integers), the number of prime factors exceeding P_0 is at most $3h \log(CN)/\log P_0$. □

Theorem 53 (Polylogarithmic exceptional set)**.** *For every $\varepsilon > 0$,*

$$\#\{n \leq N : \log G_n > \varepsilon\, n\} = O_{\varepsilon}\big((\log N)^2\big).$$

Proof. Fix $\varepsilon > 0$ and consider $n \in (N, 2N]$ with $\log G_n > \varepsilon n$. By Proposition 3, the small-prime contribution is $O(\sqrt{n})$. Each prime $p > \sqrt{n}$ satisfies $v_p(G_n) \leq 3$ (since $v_p(d_n) = 3$). By Lemma 48, leading-digit bad primes ($p \mid b_{\lfloor n/p \rfloor}$ with $n \bmod p \notin \mathcal{Z}_p$) satisfy $v_p(G_n) = 0$. The remaining primes with $v_p(G_n) \geq 1$ are the lower-digit bad primes ($n \bmod p \in \mathcal{Z}_p$) and the *companion-block primes* ($D_q \equiv 0 \pmod{p}$ with $p \nmid b_r$, cf. Lemma 47). By Proposition 51, the total logarithmic weight of companion-block primes is $O(n^{2/3})$ at every index, hence $o(n)$. Hence $\log G_n \leq O(\sqrt{n}) + 3\, t(n) \log n$, and $\log G_n > \varepsilon n$ forces

$$t(n) := \#\{p > \sqrt{n} : n \bmod p \in \mathcal{Z}_p\} \geq \frac{\varepsilon n}{4 \log n} \tag{14}$$

for N sufficiently large.

Partition $(N, 2N]$ into intervals J of length $Y = c_1 \varepsilon^2 N/\log N$, where $c_1 > 0$ is a small constant to be chosen. Write $\mathcal{E}_J = \{n \in J : \log G_n > \varepsilon n\}$ and $M = |\mathcal{E}_J|$.

For each prime $p \in (\sqrt{N}, 2N]$, let d_p count the elements of $\mathcal{E}_J$ for which p is lower-digit bad. By (14),

$$I := \sum_p d_p \geq \frac{\varepsilon N M}{4 \log N}.$$

The prime universe has size $L = \pi(2N) \le 4N/\log N$.

Upper bound on pair incidence. For $m, n \in \mathcal{E}_J$ with $m < n$, Lemma 52 gives at most $C(n-m)$ common lower-digit bad primes (taking $P_0 = N^\delta$ in the codegree lemma). Hence

$$\sum_p \binom{d_p}{2} = \sum_{\substack{m<n \\ m,n\in\mathcal{E}_J}} |\mathcal{P}_m \cap \mathcal{P}_n| \le C \sum_{\substack{m<n \\ m,n\in\mathcal{E}_J}} (n-m) \le CYM^2.$$

Lower bound on pair incidence. By Cauchy–Schwarz applied to the d_p,

$$\sum_p \binom{d_p}{2} \ge \frac{1}{2}\left(\frac{I^2}{L} - I\right) \ge \frac{1}{2}\left(\frac{\varepsilon^2 N M^2}{64\log N} - I\right).$$

Combining gives $CYM^2 \ge \varepsilon^2 NM^2/(128\log N) - I/2$. Since $I \le LM \le 4NM/\log N$ and $Y = c_1\varepsilon^2 N/\log N$, for c_1 chosen smaller than $1/(128C)$, the YM^2 term cannot absorb the quadratic lower bound once $M > C'/\varepsilon^2$. Hence $M = O(1/\varepsilon^2)$.

There are $N/Y = O(\log N/\varepsilon^2)$ intervals. The total on $(N, 2N]$ is therefore $O(\log N/\varepsilon^4)$. Summing over $O(\log N)$ dyadic blocks $(N/2^{j+1}, N/2^j]$ gives the result. □

Theorem 53 supersedes Corollary 43. The proof's critical-window estimate—every interval of length $\asymp \varepsilon^2 N/\log N$ in $(N, 2N]$ contains $O_\varepsilon(1)$ exceptional integers—directly gives the uniform local bound

$$\sup_{M\ge 0} \#\{M < n \le M + L : \log G_n > \varepsilon n\} \ll_\varepsilon L^{2/3} + 1,$$

by applying the critical-window argument to any interval $[M, M+L]$ with $N = M$ and $Y = c\varepsilon^2 M/\log M$. Dividing by L and letting $L \to \infty$ yields *upper Banach density zero* unconditionally, strengthening the harmonic-weight corollary 44.

9 Counting bad primes: the conditional result

We first sharpen the leading-digit incidence from $O(N^{3/2})$ to $O(N^{10/7}/\log N)$ via a dyadic argument combining divisor-counting with the sub-interval zero bound; this is unconditional and also strengthens the conditional theorem below.

Proposition 54 (Dyadic leading-digit bound)**.** *The leading-digit incidence L_N (counting pairs (n, p) with $n \in (N, 2N]$, $p > \sqrt{n}$ prime, and $p \mid b_{\lfloor n/p\rfloor}$) satisfies*

$$L_N = O\left(\frac{N^{10/7}}{\log N}\right).$$

Proof. For dyadic blocks $P = 2^j$, $Q = 2^k$ with $Q < P$ and $PQ \in [N/4, 4N]$, define

$$D(P, Q) = \#\{(p, q) : P < p \le 2P,\ Q < q \le 2Q,\ p \mid b_q\}.$$

Two bounds hold unconditionally:

- *Sub-interval zero counting:* By Corollary 10, each prime $p \in (P, 2P]$ has at most $O(Q^{2/3})$ zeros of b_j in $(Q, 2Q]$. Summing over $\pi(2P)$ primes gives $D(P, Q) = O(P\,Q^{2/3}/\log P)$.

- *Divisor counting:* Since $\log b_q \le (q+1)\log 64$, the number of prime factors of b_q exceeding P is at most $\log b_q/\log P = O(q/\log P)$. Summing over $q \in (Q, 2Q]$ gives $D(P,Q) \le \sum_q O(q/\log P) = O(Q^2/\log P)$.

Hence $D(P,Q) \le \min(P\,Q^{2/3},\, Q^2)/\log P$.

Each pair (p,q) in $D(P,Q)$ contributes to at most $p \le 2P$ values of n with $\lfloor n/p \rfloor = q$, so the block's contribution to L_N is at most $D(P,Q)\cdot 2P \le 2\min(P^{4/3}\,N^{2/3},\ N^2/P)/\log P$ (using $Q = \Theta(N/P)$). The crossover is at $P_0 = N^{4/7}$. Summing $O(\log N)$ dyadic blocks $P \in [\sqrt{N}, 2N]$:

$$L_N \le \sum_{P \le N^{4/7}} \frac{2P^{4/3}\,N^{2/3}}{\log P} + \sum_{P > N^{4/7}} \frac{2N^2}{P\log P} = O\left(\frac{N^{10/7}}{\log N}\right),$$

since both sums are geometric series dominated by the $P = N^{4/7}$ boundary term. $\square$

Theorem 55. *Assume Hypothesis $\bar{Z}$. Then for a set of positive integers n of natural density* 1,

$$\log G_n = O(\sqrt{n}).$$

Proof. Fix N and $\omega(N) \to \infty$ arbitrarily slowly. As in Theorem 42, decompose $\log G_n = \sum_p v_p(G_n)\log p$. The small-prime contribution is $O(\sqrt{n})$. Split $B(n) = B_{\mathrm{low}}(n) + B_{\mathrm{lead}}(n)$, where B_{low} counts bad primes $p > \sqrt{n}$ with $n \bmod p \in \mathcal{Z}_p$ and B_{lead} counts those with $\lfloor n/p \rfloor \in \mathcal{Z}_p$.

Lower digit. Under Hypothesis $\bar{Z}$, set $M(t) = \sum_{p \le t} Z(p) \ll t/\log t$. Abel summation gives $\sum_{\sqrt{N} < p \le 2N} Z(p)/p = O(1)$, so $R_N = \sum B_{\mathrm{low}}(n) = O(N)$. By Markov, $B_{\mathrm{low}}(n) \le \omega(N)$ for all but $O(N/\omega(N)) = o(N)$ values.

Leading digit. As in the proof of Corollary 57, Hypothesis $\bar{Z}$ gives $L_N = O(N^{4/3}/\log N)$. By Markov with threshold $\omega(N)\,N^{1/3}/\log N$:

$$\#\{n \in (N, 2N] : B_{\mathrm{lead}}(n) > \omega(N)\,N^{1/3}/\log N\} \le \frac{O(N^{4/3}/\log N)}{\omega(N)\,N^{1/3}/\log N} = O(N/\omega(N)) = o(N).$$

Conclusion. For all remaining n:

$$\log G_n \le O(\sqrt{n}) + 3\big(\omega(N) + \omega(N)\,N^{1/3}/\log N\big)\log n.$$

Since $\omega(N)\cdot N^{1/3} \ll N^{1/2}$ for any $\omega(N) \to \infty$ slowly enough ($\omega \le N^{1/6}$ suffices), the second term is $O(N^{1/2}) = O(\sqrt{n})$. $\square$

Remark 56 (The leading-digit bottleneck). The gap between the conditional bound $O(\sqrt{n})$ (Theorem 55) and the conjectured $O(\log n)$ comes from the small-prime contribution (Proposition 3), which gives an unconditional floor of $O(\sqrt{n})$. Under Hypothesis $\bar{Z}$, the big-prime contribution is now absorbed into this floor, thanks to Proposition 54.

Heuristically, $B_{\mathrm{lead}}(n) = O(1)$ because the leading digit $\lfloor n/p \rfloor$ lands in $\mathcal{Z}_p$ with probability $\le C/p$ for each prime, giving expected $B_{\mathrm{lead}} = \sum C/p = O(1)$. Making this rigorous would require $D(P,Q) \ll Q/\log P$ (the random-divisibility prediction), improving the proven $D(P,Q) \le \min(P\,Q^{2/3},\, Q^2)/\log P$. The small-prime bound $O(\sqrt{n})$ itself would need a finer analysis of $v_p(G_n)$ for small primes.

Corollary 57 (Conditional exceptional set)**.** *Assume Hypothesis $\bar{Z}$. For every $\varepsilon > 0$,*

$$\#\{n \le N : \log G_n > \varepsilon\, n\} = O_\varepsilon(N^{1/3}).$$

Proof. Under Hypothesis $\bar{Z}$, $R_N = O(N)$. For the leading-digit incidence, Hypothesis $\bar{Z}$ also strengthens the dyadic bound: since each prime $p \in (P, 2P]$ contributes at most $Z(p)$ zeros to any sub-interval, $D(P,Q) \le \sum_{P<p\le 2P} Z(p) = O(P/\log P)$ (using $\sum_{p\le x} Z(p) = O(\pi(x))$). The crossover with the divisor bound $D(P,Q) \le Q^2/\log P$ shifts from $P_0 = N^{4/7}$ to $P_0 = N^{2/3}$, giving $L_N = O(N^{4/3}/\log N)$. The threshold $B(n) > \varepsilon N/(7\log N)$ is exceeded by at most

$$\frac{O(N) + O(N^{4/3}/\log N)}{\varepsilon N/(7\log N)} = O(N^{1/3}/\varepsilon).$$

Dyadic summation over $(N/2^{j+1}, N/2^j]$ gives the global bound. ☐

Under Hypothesis $\bar{Z}$, the exceptional set is not only sparse globally but cannot cluster in any interval:

Corollary 58 (Upper Banach density zero)**.** *Assume Hypothesis $\bar{Z}$. For every $\varepsilon > 0$, the set $E_\varepsilon = \{n : \log G_n > \varepsilon n\}$ has upper Banach density zero:*

$$\lim_{L\to\infty} \sup_{M\ge 0} \frac{|E_\varepsilon \cap (M, M+L]|}{L} = 0.$$

Proof. For $I = (M, M+L]$ with $M > 2L$, every prime $p > n/2$ with $n \in I$ lies in $(M/2,\, M+L]$. Under Hypothesis $\bar{Z}$, each such prime contributes at most $Z(p)$ candidate bad integers to I, giving $\sum_{n\in I} B(n) \le \sum_{M/2<p\le M+L} Z(p) \ll \pi(M+L) = O(M/\log M)$. For $n \in I \cap E_\varepsilon$, the bound $\log G_n \le O(\sqrt{n}) + 3B(n)\log n$ and $\log G_n > \varepsilon n$ force $B(n) \gg \varepsilon M/\log M$ for M large. Therefore $|I \cap E_\varepsilon| \ll_\varepsilon 1 = o(L)$. The case $M \le 2L$ follows similarly by covering $(0, 3L]$ with $O(\log L)$ dyadic intervals. ☐

Corollary 59. *The condition $Z(p) = o(p)$ is* necessary *for the conjecture: if $Z(p) \ge \varepsilon p$ for a positive proportion of primes, double-counting gives $\log G_n \ge c\varepsilon n$ for a positive proportion of n. Conversely, $Z(p) = o(p)$ implies $G_n = e^{o(n)}$ for density 1 (Theorem 42).*

Proof. For each prime $p \in (n/2, n]$ with $Z(p) \ge \varepsilon p$, Lemma 39 gives $v_p(G_n) \ge 1$ whenever $n - p \in \mathcal{Z}_p$. Suppose a proportion $\delta > 0$ of primes satisfy $Z(p) \ge \varepsilon p$. Summing over $n \in (N, 2N]$ and bad primes $p \in (n/2, n]$, the total incidence count is $\ge \delta\,\pi(2N)\cdot \varepsilon N/2 = \Omega(N^2/\log N)$. Dividing by N and applying Markov's inequality: a proportion $\ge \varepsilon\delta/4$ of integers n have $B(n) \ge \varepsilon\delta N/(4\log N)$, giving $\log G_n \ge 3B(n)\log(N/2) \ge c\varepsilon\delta\, n$. ☐

Remark 60 (The pointwise gap). The full conjecture $G_n = e^{o(n)}$ for *every* n is strictly stronger than the density-one statement. By Corollary 40, it is equivalent to the top-half radical estimate (13). The height bound gives $B(n) = O(n/\log n)$, but for an *adversarial* family (Z_p) with $|Z_p| \le 1$, one may choose $Z_p = \{N - p\}$ for primes in $(N/2, N]$, producing $B(N) \ge \pi(N) - \pi(N/2) \sim N/(2\log N)$. Thus the needed pointwise bound $B(n) = o(n/\log n)$ requires arithmetic information specific to the Apéry sequence, not just the cardinalities $Z(p)$.

By Corollary 40, every bad prime $p > n/2$ satisfies $p \mid b_n$. Hence the conjecture implies $\log \operatorname{rad}_{(n/2,n]}(b_n) = o(n)$: the large prime radical of b_n must be subexponential, even though $|b_n| = e^{\Theta(n)}$.

The Poisson model predicts that $B(n) \ge 1$ occurs infinitely often: $\mathbb{P}(B(n) \ge 1) \sim \log 2/\log n$, and $\sum(\log n)^{-1}$ diverges. This is compatible with the conjecture, since one bad prime contributes only $O(\log n)$ to $\log G_n$. To violate $G_n = e^{o(n)}$, one needs $B(n) \ge \varepsilon n/(6\log n)$; the Poisson tail gives $\mathbb{P}(B(n) \ge k_n) \le e^{-cn}$ for $k_n = \lceil \varepsilon n/(6\log n)\rceil$, and the sum converges. Thus the model predicts: bad-prime occurrences are inevitable, but catastrophic pile-ups are finitely many.

10 Computational verification

We computed G_n exactly for $n = 1, \ldots, 1000$ using rational arithmetic.

n range	avg $\log(G_n)/n$	max $\log(G_n)/n$
1–50	0.276	0.93
51–100	0.108	0.26
101–200	0.086	0.18
201–300	0.048	0.10
301–400	0.038	0.08
401–500	0.024	0.05
501–600	0.029	0.06
601–700	0.017	0.04
701–800	0.014	0.04
801–900	0.016	0.04
901–1000	0.013	0.03

A power-law fit gives $\log(G_n)/n \sim 5\,n^{-0.87}$, i.e., $\log G_n \sim 5\,n^{0.13}$, far stronger than the density-one bound $o(n)$ of Theorem 42.

The exponent 0.13 should be interpreted with caution: if $\log G_n \sim \kappa \log n$, the effective log-log slope is $1/\log n = 0.145$ at $n = 1000$, close to the fitted 0.13. A random-gcd model—assigning each prime $p \le n$ an independent probability $\sim 1/p$ of dividing G_n—predicts $\mathbb{E}[\log G_n] \sim \log n$ by the weighted Mertens theorem, with standard deviation $\sim (\log n)/\sqrt{2}$ of the same order. The data thus cannot yet distinguish $\kappa \log n$ from $n^{0.13}$; the logarithmic prediction is arithmetically more natural.

11 Remarks

1. The Beukers supercongruence $b_{mp^r} \equiv b_{mp^{r-1}} \pmod{p^{3r}}$ [4] provides tower rigidity: for each fixed prime, $v_p(b_n)$ is controlled by the base-p digits. Quantitatively, since $b_n \equiv \prod b_{d_i} \pmod{p}$ (Theorem 4), we have $p \nmid b_n$ iff every base-p digit of n avoids $\mathcal{Z}_p$. Hence

$$\#\{n < p^r : p \mid b_n\} = p^r - (p - Z(p))^r.$$

2. For the gap-2 polynomial $C_2(m) = P(m+1)$, the factorization $P(x) = (2x+1)(17x^2+17x+5)$ shows that gap-2 pairs can only occur at $m = (p-3)/2$ (when $2(m+1)+1 = 2m+3 \equiv 0$) or at roots of the quadratic $17x^2 + 17x + 5$ (discriminant -51; exists over $\mathbb{F}_p$ iff $\left(\frac{-51}{p}\right) = 1$). Empirically, all observed gap-2 pairs arise from the linear factor at $m = (p-3)/2$.

3. For $1 \le j \le p-2$, the Apéry number $b_j \bmod p$ admits a character-sum representation as a Greene [12] Gaussian hypergeometric ${}_4F_3$ over $\mathbb{F}_p$ with parameter $\chi = \omega^j$ (Teichmüller character). However, the Weil bound controls the archimedean size $|b_j|_\infty = O(p^{3/2})$ but does *not* bound $Z(p)$, since zeros of b_j modulo p are a p-adic event. The gap polynomial argument above is specifically adapted to this p-adic zero-count problem.

4. Malik–Straub [15] studied Lucas congruences and zero-divisibility for all sporadic Apéry-like sequences; Rowland–Yassawi [18] proved automaticity of the Lucas property for diagonals of rational functions, which includes the Apéry sequence. Our $Z(p)$ data for the $\zeta(3)$ sequence agrees with their empirical findings.

5. In the standard toric coordinate, the truncated polynomial $H_p(t) = \sum_{j=0}^{p-1} b_j\, t^j \in \mathbb{F}_p[t]$ is the scalar Hasse–Witt invariant of the Apéry K3 pencil. Its *roots* in $\mathbb{F}_p$ are the nonordinary fibers; because the family has generic Picard rank 19, nonordinary implies supersingular at good primes. Our statistic $Z(p)$ counts vanishing *coefficients* of H_p, equivalently the vanishing finite Mellin coefficients $b_j = -\sum_{t\in\mathbb{F}_p^\times} H_p(t)\, t^{-j}$ for $1 \le j \le p-2$. Thus $Z(p)$ is a spectral sparsity statistic for the global Hasse/point-count function, *not* a count of supersingular fibers. Standard Hasse-stratification, Newton-polygon, and overconvergent F-crystal theory do not by themselves bound $Z(p)$; they control the root locus, not the Fourier support. A bounded-complexity geometric lift *does* exist: the lisse sheaf $\mathcal{V}_\ell = R^2 f_* \mathbb{Q}_\ell$ on the smooth locus of the K3 pencil (rank 22, conductor independent of p) satisfies $\mathrm{Tr}(\mathrm{Frob}_t \mid \mathcal{V}_\ell) \equiv A_p(t) \pmod p$ by the Hasse–Witt congruence. The Mellin sum $M_{p,j} = \sum_t \tau_p(t)\, \omega_p(t)^{-j}$ is then realized on $R\Gamma_c(U_{\bar{\mathbb{F}}_p}, \mathcal{V}_\ell \otimes \mathcal{L}_{\omega_p^{-j}})$ with bounded rank and conductor. However, the event $b_j \equiv 0 \pmod p$ is the *same-characteristic* divisibility $\mathfrak{p}_p \mid M_{p,j}$, a crystalline condition outside the scope of ℓ-adic equidistribution (Forey–Fresán–Kowalski [11]). The missing theorem is a *crystalline Mellin anti-concentration* estimate controlling $\#\{j : p \mid M_{p,j}\}$.

6. **Palindromic symmetry.** For $0 \le j \le p-1$ and $0 \le k \le p-1-j$, the identity $\binom{p-1-j}{k} \equiv (-1)^k \binom{j+k}{k} \pmod p$ holds (each of the k falling factors $p-1-j-i \equiv -(j+1+i)$). When $k \le j$, $\binom{p-1-j+k}{k} \equiv (-1)^k \binom{j}{k}$; when $k > j$, Lucas's theorem gives $\binom{p-1-j+k}{k} \equiv 0$ (since $p-1-j+k \ge p$ and $\binom{k-j-1}{k} = 0$). Therefore each term of $b_{p-1-j} = \sum_k \binom{p-1-j}{k}^2 \binom{p-1-j+k}{k}^2$ with $k > j$ vanishes, and for $k \le j$ the $(-1)^{2k}$ signs cancel, giving $b_{p-1-j} \equiv b_j \pmod p$. The extra terms of b_j when $j > (p-1)/2$ (i.e., $k > p-1-j$) likewise vanish: $\binom{j+k}{k} \equiv \binom{j+k-p}{k} \equiv 0$ since $j+k-p < k$. Equivalently, $H_p(t)$ is palindromic: $t^{p-1} H_p(1/t) \equiv H_p(t)$. This explains the observed pairing: whenever $b_j \equiv 0$, also $b_{p-1-j} \equiv 0$, so zeros of b_j come in pairs $\{j, p-1-j\}$ (except at the self-paired center $j = (p-1)/2$, which vanishes only at non-ordinary primes). Proposition 50 proves the stronger statement that *every* solution of the Apéry recurrence is palindromic modulo p, in particular the companion solution a. The "pair count" $K(p) = Z(p)/2$ (or $(Z(p)-1)/2$ at non-ordinary primes) follows Poisson(1/2) to the precision of the data (Proposition 28).

7. **Sym^2 squareness (theorem).** The Apéry differential operator is the symmetric square of the Picard–Fuchs operator of an underlying elliptic family (Stienstra–Beukers [20]). Caruso, Fürnsinn, Vargas-Montoya, and Zudilin [7] proved the mod-p squareness rigorously: for every prime $p \ge 5$, setting $\varepsilon_p = \frac{1}{2}(1 - (\frac{-6}{p}))$, there exists a unique (up to sign) polynomial $B_p \in \mathbb{F}_p[t]$ such that
$$H_p(t) \;=\; \Delta(t)^{\varepsilon_p} \cdot B_p(t)^2,$$
where $\Delta(t) = t^2 - 34t + 1$ is the conifold discriminant and B_p is squarefree and coprime to Δ. The proof uses the explicit identity $F(t(x)) = (1+x)\, h(x)^2$ between the Apéry and Franel generating functions, the p-Lucas property of both sequences, and a quadratic-descent argument through the covering $\mathbb{F}_p(x)/\mathbb{F}_p(t)$ where $t = x(1-8x)/(1+x)$. The deck-transformation character $(-6/p)$ determines whether $\varepsilon_p = 0$ or 1:
$$\varepsilon_p = 0 \iff \left(\frac{-6}{p}\right) = 1, \qquad \varepsilon_p = 1 \iff \left(\frac{-6}{p}\right) = -1.$$
Equivalently, $\varepsilon_p = 0$ iff $p \equiv 1, 5, 7, 11 \pmod{24}$. These two cases occur with equal density 1/2 by Dirichlet's theorem.

Splitting the mean $Z(p)$ by residue class modulo 8 (and modulo 24) yields no significant difference (mean ≈ 1.00 in each class, $n \geq 2000$ primes per class up to 10^5); the squareness structure does not directly influence $Z(p)$, consistent with $Z(p)$ counting coefficient zeros rather than root zeros.

The factorization imposes a convolution structure: b_j is (up to the Δ^{ε_p} correction) the j-th coefficient of B_p^2, i.e., $b_j \equiv \sum_k \beta_k \beta_{j-k} \pmod p$ where β_k denotes the coefficients of B_p. The vanishing $b_j \equiv 0$ thus requires cancellation across $\sim p/2$ convolution terms. A Monte Carlo simulation (1000 random monic polynomials of degree $(p-1)/2$ over $\mathbb{F}_p$, for $p \in \{101, 503, 1009, 5003\}$) shows that the zero-count of the *squared* polynomial follows Poisson(1) (mean ≈ 1, variance ≈ 1, total variation distance to Poisson(1) equal to 0.018). However, the observed $Z(p)$ distribution over 667 primes $p \leq 5000$ is *not* Poisson(1) (TV = 0.43); instead it matches $2 \cdot$ Poisson(1/2) with TV = 0.023. The explanation is the palindromic pairing: since $b_{p-1-j} \equiv b_j$, zeros come in pairs $\{j, p-1-j\}$ and $Z(p)$ is almost always even. Denoting by $Z_{\text{pair}}(p)$ the number of zero pairs, the data show $Z_{\text{pair}} \sim$ Poisson(1/2), so $Z(p) = 2Z_{\text{pair}}(p) \sim 2 \cdot$ Poisson(1/2). Thus squareness explains the $O(1)$ mean, while palindromic symmetry supplies the parity constraint. The Poisson tail predicts $\max_{p \leq N} Z(p) \sim 2 \log\log N$ via extreme-value theory, strongly suggesting that $Z(p)$ is unbounded (i.e., Hypothesis Z is false). Whether the algebraic provenance of B_p imposes a bounded *average* (Hypothesis $\bar{Z}$) remains the central open question.

The factorization does *not* improve the divisor-counting bound $D(P, Q) \leq Q^2/\log P$. In the regime $P \geq Q^{4/3}$ where the divisor estimate is active, one has $q < p/2$, so the coefficient b_q mod p lies in the formally free triangular range of the square root: the map $(\beta_1, \ldots, \beta_q) \mapsto (b_1, \ldots, b_q)$ mod p is a triangular automorphism with Jacobian 2^q, hence the single equation $b_q \equiv 0$ imposes no algebraic constraint beyond fixing one coefficient of B_p. Thus the Sym^2 structure gives the correct *random model* (uniform square root $\Rightarrow$ Poisson(1/2)) but not the missing *distribution theorem* for the deterministic Apéry square root.

8. **Toric framework.** The Laurent polynomial $f = w(x + x^{-1} + y + y^{-1} + z + z^{-1})$ on $(\mathbb{F}_p^\times)^4$ has Newton polytope $\Delta = \mathrm{conv}(\{0\} \cup \mathrm{Supp}(f))$, a 4-dimensional pyramid over the regular octahedron in $\mathbb{R}^3$. Its f-vector is $(7, 18, 20, 9)$, its normalized volume is $4! \cdot \mathrm{vol}(\Delta) = 8$, and it is Kouchnirenko nondegenerate for every prime $p \geq 5$: for each of the 27 nonempty faces σ not containing the origin, the toric critical system $\{x_i \partial f_\sigma / \partial x_i = 0\}$ has no solution in $(\mathbb{F}_p^\times)^4$. For the Adolphson–Sperber twisted Hodge polygon with w-Kummer twist $\chi_w = \omega^{-j}$ (character parameter $a = j/(p-1)$), the slopes are a, $a+1$, $a+2$, $a+3$ with multiplicities $1, 3, 3, 1$, confirming that the first slope equals a for every $a \in \{k/100 : 0 \leq k \leq 99\}$. By Newton-above-Hodge (Adolphson–Sperber [2]), the p-adic Newton polygon lies on or above this Hodge polygon, and ordinarity (Newton = Hodge) implies the existence of a unit root. The $Z(p)$ bound thus reduces to proving ordinarity for all but $O(1)$ twist parameters j.

9. **Branch obstruction.** The $p-1$ Kummer twists χ^{-j} $(j = 0, \ldots, p-2)$ are distinct characters of conductor p. In Iwasawa-theoretic coordinates $\chi^j = \omega^a \langle\cdot\rangle^s$ with $a = j \bmod (p-1)$, the horizontal family $\{j : 0 \leq j < p-1\}$ hits *each* of the $p-1$ branches ω^a exactly once. Weierstrass preparation and bounded-λ rigidity control zeros *within one branch* (the vertical direction j, $j + (p-1)$, $j + (p-1)p, \ldots$, where Dwork congruences live) but not *across* branches. Thus there is no p-adic analytic function of one variable whose zero count majorizes $Z(p)$. Any "rigidity" proof of Hypothesis $\bar{Z}$ would need a mechanism not supplied by existing Iwasawa–Dwork theory.

10. Any estimate $\log G_n = o(n)$—whether $O(\log n)$, $O(\sqrt{n})$, or $O(n^{0.13})$—yields $\gamma = 0$ in the exponent $G_n = e^{\gamma n + o(n)}$ and therefore does not improve the Rhin–Viola irrationality measure $\mu(\zeta(3)) \leq 5.513$ [19], which depends only on the exponential growth rate $|b_n\zeta(3) - a_n| = e^{-(\log\alpha + o(1))n}$. An improvement would require an exponentially large common divisor, which is not predicted by any current model.

11. **Comparison with previous work.** Classical refinements of Apéry's method (Beukers [4], Rhin–Viola [19], Zudilin [21]) exhibit explicit common factors of the hypergeometric coefficients to improve denominator estimates and irrationality measures, but always uniformly for every index. They do not provide exceptional-set estimates for the *actual* reduced gcd G_n. A separate body of work, based on the Subspace Theorem, proves $\gcd(a^n - 1, b^n - 1) = e^{o(n)}$ pointwise for multiplicatively independent a, b (Bugeaud–Corvaja–Zannier [6]); these results apply to constant-coefficient linear recurrences via Binet representations, but the Apéry recurrence's polynomially varying coefficients preclude the multiplicative-independence hypothesis. To our knowledge, no previous work gives a quantitative power-saving exceptional-set estimate for gcd cancellation in a variable-coefficient holonomic recurrence.

12. **The remaining open problem.** The companion-block height bound (Proposition 51) shows that primes $p > \sqrt{n}$ with $D_q \equiv 0 \pmod{p}$ contribute $O(n^{2/3})$ to $\log G_n$ at *every* index. After removing this channel, the conjecture $G_n = e^{o(n)}$ for all n is equivalent to the *pointwise radical estimate*

$$\sum_{\substack{p > \sqrt{n} \\ p \mid b_{n \bmod p}}} \log p \;=\; o(n).$$

Theorem 53 proves this for all but $O_\varepsilon((\log N)^2)$ integers in $[1, N]$. However, the codegree method (Lemma 52) controls primes *shared* between pairs of integers, and does not bound the total weight of "private" primes at a single index. Closing the gap requires a one-point rainbow estimate: an argument that one fixed integer cannot simultaneously be bad at too many distinct primes.

12 The second-moment approach

The codegree method of Section 8 exploits primes shared between pairs of integers to prove a density-one theorem. To close the remaining gap—proving $W(n) = o(n)$ for *every* n—we study the *second moment* of the bad-prime weight function.

12.1 The criterion

For a dyadic shell $I_N = (N, 2N]$, define

$$B(m) \;=\; \sum_{\substack{p > \sqrt{m} \\ p \mid b_{m \bmod p}}} \log p, \qquad M_2(N) \;=\; \sum_{m \in I_N} B(m)^2.$$

Proposition 61 (Second-moment criterion)**.** *If $M_2(N) \leq C \cdot N^{2-\delta}$ for some $C > 0$ and $\delta > 0$, then $W(n) = o(n)$ for all n. More precisely, for every $n \in I_N$,*

$$W(n) \;\leq\; B(n) \;\leq\; \sqrt{M_2(N)} \;\leq\; C^{1/2} \cdot N^{1-\delta/2}.$$

Proof. $B(n)^2$ is one nonneg. term of the sum $M_2(N)$, hence $B(n) \leq \sqrt{M_2(N)}$. ∎

In particular, $M_2(N) = O(N \cdot (\log N)^A)$ for any fixed A gives $B(n) = O(\sqrt{N} \cdot (\log N)^{A/2}) = o(N)$.

Remark 62 (Shifted-gcd identity). The bad-prime weight admits a combinatorial reformulation. By the injectivity of $p \mapsto n \bmod p$ on primes $p > \sqrt{n}$ (two such primes giving the same residue would force their product $\le n$),

$$W(n) \;=\; \sum_{0<r<n} \log \operatorname{rad}_{>\max(\sqrt{n},\, r)} \gcd(n-r,\, b_r).$$

Each inner radical is 1 or a single prime $p = n - r$. For the top-half block $p > n/2$, the condition becomes "$n - r$ is prime and $(n-r) \mid b_r$."

Proposition 63 (Hard-core reduction)**.** *For each bad prime $p > \sqrt{n}$, write $n = q_p p + r_p$, $s_p = \min(r_p, p-1-r_p)$, $h_p = |p-1-2r_p|$. Let $1 \le Q, R, H < n/2$. Then the total weight of primes satisfying at least one of $q_p > Q$, $s_p \le R$, $h_p \le H$ is $O(n/Q + (R+H)\log n)$. In particular, with $Q = R = H = \sqrt{n/\log n}$, all "easy channels" contribute $O(\sqrt{n \log n})$; the remaining hard core consists of primes $p \gg \sqrt{n \log n}$ with zero positions far from both endpoints and the reflection centre.*

Proof. Large quotient ($q_p > Q$): then $p < n/Q$, so $\sum \log p \le \vartheta(n/Q) \ll n/Q$. Near endpoint ($s_p \le R$): by the injectivity of $p \mapsto n \bmod p$, at most two primes $> \sqrt{n}$ contribute for each value of s_p. Near centre ($h_p \le H$): the identity $(2q_p + 1)p = 2n + 1 \pm h_p$ forces p to divide a fixed integer of size $O(n)$, giving $O(1)$ primes per value of h_p. □

Remark 64 (The vertical main term is harmless). Write $W(n) = M(n) + \mathcal{E}(n)$ where $M(n) = \sum_{p>\sqrt{n}} Z(p)\,(\log p)/p$. The unconditional bound $Z(p) \le Cp^{2/3}$ gives $M(n) \ll \sum_{p\le n} p^{-1/3} \log p \ll n^{2/3}$. The entire problem is the oscillatory error $\mathcal{E}(n)$, which at each prime is $(\log p) \cdot (1_{n \bmod p \in \mathcal{Z}_p} - Z(p)/p)$.

12.2 Empirical verification

The following table, computed for all dyadic shells with $64 \le N \le 4096$ using the full zero-set data for all primes $p \le 20{,}000$, gives:

N	$M_2(N)$	M_2/N	$M_2/(N\log^2 N)$	$\max B(m)$
64	672	10.5	0.605	10.4
128	1,684	13.2	0.560	15.9
256	4,376	17.1	0.555	15.8
512	11,896	23.2	0.598	22.3
1024	28,632	28.0	0.583	21.7
2048	69,226	33.8	0.582	27.1
4096	166,671	40.7	0.587	32.1
8192	421,868	51.5	0.634	37.2

The ratio $M_2(N)/(N \log^2 N)$ stabilizes near 0.59, strongly suggesting

$$M_2(N) \;\sim\; C_2 \cdot N \cdot (\log N)^2, \qquad C_2 \approx 0.59. \tag{15}$$

This gives $B(n) \le \sqrt{0.59\,N}\,\log N \approx 0.77\,\sqrt{N}\,\log N = o(N)$.

12.3 CRT expansion

Expand M_2 dyadically. For distinct primes $p, q \in (P, 2P]$ with $P > \sqrt{N}$, write $\Omega_p = \{m \in I_N : m \bmod p \in \mathcal{Z}_p\}$. Then

$$M_2(N) \;=\; \underbrace{\sum_p (\log p)^2\,|\Omega_p|}_{\text{diagonal}} \;+\; \underbrace{\sum_{p \neq q} \log p\,\log q\,|\Omega_p \cap \Omega_q|}_{\text{off-diagonal}}.$$

Diagonal. $|\Omega_p| = Z(p) \cdot N/p + O(Z(p))$, so

$$\text{diag} \;=\; N \sum_p \frac{Z(p)\,(\log p)^2}{p} \;+\; O\Big(\sum_p Z(p)\,(\log p)^2\Big).$$

Under the Poisson model $\mathbb{E}[Z(p)] = 1$, the main term is $\sim N \cdot \sum_{P<p\le 2P} (\log p)^2/p \sim N \cdot (\log P)^2 \cdot \log 2$. Summing dyadically $P = \sqrt{N}, 2\sqrt{N}, \ldots, N$ gives diagonal $\sim \frac{1}{2}\,N\,(\log N)^2$, matching $\approx 63\%$ of the empirical (15).

Off-diagonal. Since $p, q > \sqrt{N}$ we have $pq > N$, so each CRT class $r_p \pmod p$, $r_q \pmod q$ contains at most one element of I_N. The CRT main term of the off-diagonal is $\sim N\big(\sum Z(p) \log p/p\big)^2 \sim \frac{1}{4}\,N\,(\log N)^2$.

The predicted constant. Under the independent random model—each prime p hits a uniform m with probability $\sim 1/p$, independently across primes—the second moment of $B(m)$ at one integer is $\mathbb{E}[B(m)^2] = (\mathbb{E}[B])^2 + \mathrm{Var}(B)$. By weighted Mertens, $\mathbb{E}[B] = \sum_{\sqrt{m}<p\le m} (\log p)/p = \frac{1}{2}\log m + O(1)$, and

$$\mathrm{Var}(B) \;=\; \sum_{\sqrt{m}<p\le m} \frac{(\log p)^2}{p}\left(1 - \frac{1}{p}\right) \;=\; \tfrac{3}{8}\,(\log m)^2 + O(\log m).$$

Therefore

$$\mathbb{E}[B(m)^2] \;=\; \big(\tfrac{1}{4} + \tfrac{3}{8}\big)(\log m)^2 + O(\log m) \;=\; \tfrac{5}{8}\,(\log m)^2 + O(\log m), \tag{16}$$

predicting $M_2(N) \sim \frac{5}{8}\,N\,(\log N)^2$. The diagonal constant is $3/8$ (matching the observed $\mathrm{Diag}/N\log^2 N \approx 0.37$) and the off-diagonal contributes $1/4$. The empirical ratio $M_2/(N\log^2 N) \approx 0.59$ is consistent with convergence to $5/8 = 0.625$, with lower-order boundary effects.

Centered variance and cross-prime covariance. The centered dispersion $V^\circ = M_2 - M_1^2/N$ decomposes as

$$V^\circ \;=\; \underbrace{\sum_p (\log p)^2\,|\Omega_p|\big(1 - |\Omega_p|/N\big)}_{\le\,\text{Diag}} \;+\; \underbrace{\sum_{p\neq q} \log p\,\log q\,\big(|\Omega_p \cap \Omega_q| - |\Omega_p|\,|\Omega_q|/N\big)}_{E_N^\circ}. \tag{17}$$

The first sum is at most the diagonal S (since $|\Omega_p|/N \ge 0$). The term E_N° is the centered cross-prime covariance: it vanishes exactly when the events $\{m \in \Omega_p\}$ are pairwise uncorrelated. The computed values are:

N	$M_2/N\log^2 N$	$\mathrm{Diag}/N\log^2 N$	V°/S	E_N°
256	0.556	0.359	1.014	+110
512	0.597	0.376	1.010	+215
1024	0.582	0.375	0.981	−152
2048	0.581	0.370	0.992	+7
4096	0.588	0.371	0.995	+41
8192	0.634	0.387	0.995	−188
16384	0.669	0.400	0.995	−922
32768	0.646	0.394	0.988	−14065

The ratio V°/S converges to 1 from below for $N \geq 1024$, with $V^\circ/S < 1$ at every scale tested beyond 512. The negative sign of E_N° for large N reflects genuine *anti-correlation* among the CRT-constrained incidence events: at $N = 1024$, over 96% of cross-prime covariance pairs $C_{p,q}$ are negative. This is consistent with a weak form of negative dependence arising from the partial-period CRT constraint. The magnitude $|E_N^\circ|$ is $O(\sqrt{N}\,(\log N)^2)$, so $|E_N^\circ|/S \to 0$ at rate $\sim 1/\sqrt{N}$.

Three-model comparison. To test whether the observed $V^\circ/S \approx 1$ is an Apéry-specific phenomenon or a generic consequence of the cardinality and reflection structure, we compare three models on each dyadic block I_N: (1) the actual Apéry zero-sets; (2) uniformly random reflected subsets of $\mathbb{F}_p^*$ with the same cardinalities $|Z_p|$; (3) random reflected subsets with cardinalities drawn from $2 \cdot \mathrm{Poisson}(1/2)$. Over 20 independent trials of each random model and $N \in \{256, 512, 1024, 2048\}$, all three models yield $V^\circ/S = 1.00 \pm 0.05$, with no statistically significant difference between the Apéry data and either random model. This rules out any detectable cross-prime alignment or repulsion specific to the Apéry sequence.

Remark 65 (Boundary correction). The naïve CRT main term $Z(p)Z(q)N/(pq)$ overestimates the random prediction for primes near $\sqrt{N}$ or N, because the admissible interval $J_{p,q} = J_p \cap J_q$ may be shorter than N. After replacing N by $L_{p,q} = |J_{p,q}|$ in the main term, the boundary-corrected CRT residual is positive and $O(\sqrt{N}\,(\log N)^2)$—the apparent negative sign in the uncorrected decomposition is entirely a boundary artifact.

12.4 The no-go model

Define the *adversarial zero-sets*: for one fixed $m_0 \in I_N$, set $\mathcal{Z}_p^* = \{m_0 \bmod p,\ p-1-(m_0 \bmod p)\}$. Then $|\mathcal{Z}_p^*| = 2$ (satisfying reflection), yet $m_0 \in \Omega_p^*$ for *every* prime $p > \sqrt{N}$, giving $B^*(m_0) = \sum_{p>\sqrt{N}} \log p = n - o(n)$. In particular:

- The adversarial model satisfies $Z^*(p) = 2$, $Z^*_{\mathrm{pair}}(p) = 1$, and all proved vertical bounds.
- Yet $M_2^*(N) \geq B^*(m_0)^2 \geq c\,n^2$, which is $\Omega(N^2)$, not $O(N)$.

Conclusion: *No proof using only vertical information (the sizes $Z(p)$, the reflection symmetry, nearest-neighbor exclusion, and gap-polynomial bounds) can establish $W(n) = o(n)$ for all n.* The missing theorem must use cross-prime *horizontal* information: the locations of the zeros $\mathcal{Z}_p$ across different primes p.

12.5 The Fourier form of the CRT error

Define the modular Fourier transform of $\mathcal{Z}_p$:

$$S_p(a) \;=\; \frac{1}{p} \sum_{r \in \mathcal{Z}_p} e\!\left(\frac{ar}{p}\right).$$

Then $S_p(0) = Z(p)/p$ and, by the palindromic pairing, $e(a(p-1)/(2p))\, S_p(a)$ is real-valued for every a. Fourier inversion gives

$$E_N(p,q) \;=\; \sum_{\substack{(a,b)\neq(0,0)\\ 0\le a<p,\ 0\le b<q}} S_p(a)\, S_q(b) \sum_{m\in I_N} e\!\left(\frac{am}{p}+\frac{bm}{q}\right).$$

The inner sum is a standard exponential sum $|K_N(\alpha)| \le \min(N, 1/\|\alpha\|)$ where $\alpha = a/p + b/q$.

Applying the classical additive large sieve to bound $\sum_{p\neq q} |E_N(p,q)|$ gives $O(N^2/\log N)$ at the top prime scale $p \sim N$—one full logarithm larger than the threshold $o(N^2/\log^2 N)$ required for $M_2 = O(N(\log N)^A)$. Thus the classical large sieve does not close the gap.

12.6 The dispersion target

Define the centered dispersion

$$V^\circ(P,N) \;=\; \sum_{m\in I_N} \left| B_P(m) - \frac{S(P,N)}{N} \right|^2,$$

where $B_P(m) = \sum_{P<p\le 2P} 1_{\Omega_p}(m)$ and $S(P,N) = \sum_p |\Omega_p|$.

Hypothesis 66 (AP–BDH dispersion)**.** $V^\circ(P,N) \ll N^{o(1)} \cdot S(P,N)$.

Proposition 67. *Hypothesis 66 together with the unconditional bound $Z(p) = O(p^{2/3})$ implies $W(n) = o(n)$ for all n.*

Proof. By hypothesis, for every $m \in I_N$,

$$B_P(m) \;\le\; \frac{S(P,N)}{N} + \sqrt{V^\circ(P,N)} \;\le\; \frac{S(P,N)}{N} + N^{o(1)}\sqrt{S(P,N)}.$$

By the zero-count bound, $S(P,N) \le (N/P+1)\cdot \sum_{P<p\le 2P} Z(p) \ll NP^{2/3}/\log P$. Therefore $B_P(m) \ll N^{o(1)} \cdot (NP^{2/3}/\log P)^{1/2}$. The weighted sum satisfies $W_P(m) \le B_P(m)\cdot \log(2P)$. Summing over $O(\log N)$ dyadic blocks gives

$$W(n) \;\ll\; N^{o(1)} \cdot N^{1/2} \cdot N^{1/3} = N^{5/6+o(1)} = o(N). \qquad \square$$

The dispersion hypothesis is a cross-prime equidistribution theorem stating that the Apéry zero-sets $\mathcal{Z}_p$ do not systematically align at one index across different primes. The no-go model of Section 12.4 shows that this hypothesis genuinely requires arithmetic input about the Apéry family, beyond what any vertical bound can provide.

Remark 68 (Covariance sign distribution). At $N = 1024$, more than 93% of the $\binom{113}{2} = 6{,}328$ cross-prime covariance terms $C_{p,q}$ are negative. This negativity has a clean structural explanation: for primes $p, q > \sqrt{N}$, the Poisson parameter $\lambda_{p,q} = A_pA_q/N < 1$, so $C_{p,q} < 0$ precisely when the joint count $J_{p,q}$ vanishes. In the sparse regime, $J_{p,q} = 0$ for $e^{-\lambda} \approx 96\%$ of pairs, while the rare pairs with $J_{p,q} = 1$ have $C_{p,q} = 1 - \lambda_{p,q} \approx 1$. Over a complete period $\pmod{pq}$, the average of $C_{p,q}$ is exactly zero: many small negatives are balanced by few large positives. Consequently, the high percentage of negative signs does *not* imply negative dependence and cannot be used directly. The aggregate off-diagonal $E^\circ = \sum_{p\neq q} w_pw_qC_{p,q}$ oscillates in sign: positive at $N = 1024, 4096, 8192$ and negative at $N = 2048$ (Table 1). The relevant theorem is the aggregate *sub-Poisson collision* estimate

$|E^\circ| = o(S)$, which requires cancellation in the signed sum rather than a universal sign or a factored negative-dependence property. Neither the Dubhashi–Ranjan negative-lattice theorem (designed for balls-into-bins, not partial-period CRT) nor Janson's inequality (which requires nonnegativity of intersection parameters) applies to this partial-period setting.

Remark 69 (The dispersion route bypasses $\bar{Z}$). Hypothesis 66 does *not* require Hypothesis $\bar{Z}$. The adversarial singleton model $\mathcal{Z}_p^* = \{m_0 \bmod p, p{-}1{-}(m_0 \bmod p)\}$ satisfies $\bar{Z}$ in its strongest form ($Z(p) = 2$ for all p), yet violates the dispersion hypothesis. Conversely, the dispersion hypothesis with $Z(p) = O(p^{2/3})$ already gives $W(n) = o(n)$, even without any average bound on $\sum Z(p)$. This makes the dispersion route *strictly stronger* than the $\bar{Z}$ route: it is both sufficient on its own and robust against arbitrary zero-count fluctuations. The three-model comparison confirms that $V^\circ/S \approx 1$ empirically, matching the prediction of the independent random model.

Remark 70 (Martingale decomposition). Order the active primes as $p_1 < \cdots < p_K$ and define the Doob martingale $M_k = \mathbb{E}[B \mid X_{p_1}, \ldots, X_{p_k}]$. The increments $D_k = M_k - M_{k-1}$ are orthogonal: $\mathrm{Var}(B) = \sum_k \mathbb{E}[D_k^2]$. However, $D_k = (X_{p_k} - \pi_k)\Delta_k$ where the "predictive cascade" $\Delta_k = w_{p_k} + \sum_{j>k} w_{p_j}(\mathbb{P}(X_{p_j}{=}1 \mid \mathcal{F}_{k-1}, X_{p_k}{=}1) - \mathbb{P}(X_{p_j}{=}1 \mid \mathcal{F}_{k-1}, X_{p_k}{=}0))$ contains all future primes. Under independence $\Delta_k = w_{p_k}$ and the variance equals the diagonal. For the Apéry measure on a partial CRT period, revealing that $m \bmod p_k$ falls in a zero can nearly identify m (when $A_{p_k} \approx 1$), making most subsequent indicators nearly deterministic. Controlling the cascade $\sum \mathbb{E}[\pi_k(1 - \pi_k)\Delta_k^2] \le (1 + o(1)) \sum w_{p_k}^2 \alpha_{p_k}$ is equivalent to the dispersion inequality itself.

Remark 71 (Concentration and the all-n gap). The L^2-to-uniform upgrade (Lemma 73) gives $\max |B(m) - \mu| \le \sqrt{V^\circ}$. If $V^\circ \le C\,S$ for some constant C, then $\sqrt{V^\circ} = O(\sqrt{N}\,(\log N)^{3/2}) = o(N/\log N)$, so the all-$n$ theorem $W(n) = o(n)$ follows without any higher-moment, sub-Gaussian, or negative-association input (Corollary 74). Empirically, the actual concentration is much stronger: $\max B(m)/(N/\log N) \approx 0.07$ at $N = 4096$ (with $\max B \approx 36$ vs. $N/\log N \approx 492$), and the maximum $|B(m) - \mu|/\sigma$ stabilizes near 6–7 standard deviations, consistent with the Poisson prediction for the extremal order statistic.

Remark 72 (Block decomposition of V°). Decompose the prime sum $B(m) = \sum_j B_{P_j}(m)$ by dyadic prime blocks $P_j < p \le 2P_j$. Then $V^\circ = \sum_j V_{P_j}^\circ + 2\sum_{j<k} \mathrm{cross}(P_j, P_k)$. At $N = 4096$, the cross-block covariance is 0.03% of the full V°. This is because cross-block pairs (p, q) with $p \in (P, 2P]$, $q \in (Q, 2Q]$, $Q \ge 2P$ satisfy $pq \ge 2P^2 > 2N$, so the CRT approximation is even tighter for cross-block than within-block terms. Consequently, the dispersion hypothesis reduces to $V_P^\circ \ll N^{o(1)}\,S_P$ for each dyadic block *separately*—a within-scale CRT independence statement.

12.7 The covariance structure and the L^2-to-uniform upgrade

N	K	V°/S	E°	$\lvert E^\circ\rvert/S$	$J{=}0$	$J{=}1$
512	67	0.998	+130.6	0.015	95.6%	4.1%
1024	113	0.998	+215.6	0.010	96.6%	2.9%
2048	205	0.994	+96.2	0.002	98.1%	1.8%
4096	380	0.996	+120.0	0.001	98.9%	1.1%
8192	742	1.001	+1344	0.005	99.3%	0.6%

Table 1: Covariance structure of the off-diagonal sum $E^\circ = V^\circ - S_c$, computed over all $\binom{K}{2}$ pairs of active primes $p > \sqrt{N}$. $J{=}0$ and $J{=}1$ are the fractions of pairs (p, q) with joint occupancy $J_{p,q} = 0$ or 1. The Cauchy–Schwarz bound on $|E^\circ|$ exceeds the actual value by a factor of 3,000–6,000.

The following deterministic lemma is the bridge from the dispersion hypothesis to the all-index conclusion.

Lemma 73 (L^2-to-uniform upgrade)**.** *Let $F_N \colon I_N \to \mathbb{R}$ with $\mu_N = \frac{1}{N}\sum_{m\in I_N} F_N(m)$ and $\mathcal{V}_N = \sum_{m\in I_N}(F_N(m)-\mu_N)^2$. Then*

$$\max_{m\in I_N} |F_N(m)-\mu_N| \;\le\; \sqrt{\mathcal{V}_N}. \tag{18}$$

Proof. Each squared deviation $(F_N(m_0)-\mu_N)^2$ is a single summand of $\mathcal{V}_N$, hence at most $\mathcal{V}_N$. □

Corollary 74 (Dispersion implies the all-index Apéry bound)**.** *If there exists a constant C such that $V_w^\circ(N) \le C\, S_w(N)$ for all sufficiently large dyadic N, where*

$$V_w^\circ(N) = \sum_{m\in I_N} (B(m)-\mu)^2, \qquad S_w(N) = \sum_{p>\sqrt{N}} (\log p)^2 A_p,$$

then $W(n) = o(n)$ for all n, and consequently $G_n = e^{o(n)}$ for every positive integer n.

Proof. By Lemma 73 with $F_N = B$,

$$\max_{m\in I_N} |B(m)-\mu| \;\le\; \sqrt{V_w^\circ} \;\le\; \sqrt{C\,S_w}.$$

Since $S_w \le (\log 2N)^2 \sum A_p$ and $\sum A_p = O(N\log N)$ (from the bound $Z(p) \le Cp^{2/3}$ or even $\sum Z(p) \ll \pi(x)$), we have $S_w = O(N(\log N)^3)$. Thus $\sqrt{V_w^\circ} = O(\sqrt{N}\,(\log N)^{3/2})$. The mean satisfies $\mu = O((\log N)^2)$. Therefore

$$W(n) \;\le\; B(n) \;\le\; \mu + \sqrt{V_w^\circ} = O(\sqrt{N}\,(\log N)^{3/2}) = o(N/\log N),$$

which gives $W(n) = o(n)$ upon taking $N = n$. □

Table 1 and the extended computation of V°/S through $N = 32{,}768$ (Section 10) provide strong evidence for the bound $V^\circ \le C\cdot S$ with $C \le 1.01$. The ratio $|E^\circ|/S$ decreases from 0.015 at $N = 512$ to 0.001 at $N = 4096$; the brief increase at $N = 8192$ is consistent with fluctuations of $O(1/\sqrt{N})$. No concentration inequality, negative-association theorem, or fourth-moment bound is needed: the L^2-to-L^∞ trivial inequality together with $V^\circ = O(S)$ completes the proof.

12.8 The exact random CRT model

The independent fixed-margin CRT model gives a rigorous null prediction for the covariance structure. For each active prime p, retain the observed column size A_p and choose the incidence set Ω_p uniformly among all A_p-subsets of I_N, independently across primes. Put $D_p = A_p(1-A_p/N)$ and $D_w = \sum_p w_p^2 D_p = S_{\text{diag}}$.

Proposition 75 (Random CRT second moment)**.** *Under the independent fixed-margin model, $\mathbf{E}\, C_{p,q} = 0$ and $\operatorname{Var}(C_{p,q}) = D_pD_q/(N-1)$. Moreover, $\mathbf{E}(C_{p,q}C_{p,r}) = 0$ for all triples of distinct primes. Consequently,*

$$\mathbf{E}\big[(E^\circ)^2\big] \;=\; \frac{2}{N-1}\Big(D_w^2 - \sum_p w_p^4 D_p^2\Big). \tag{19}$$

If no single prime column carries a positive proportion of D_w, then

$$\frac{\operatorname{sd}(E^\circ)}{S} \;\sim\; \sqrt{\frac{2}{N}}, \tag{20}$$

matching the empirical ratio $|E^\circ|/S$ in Table 1 to within the expected fluctuations.

Proof. Given Ω_p, the intersection count $J_{p,q}$ is hypergeometric with parameters (N, A_p, A_q), giving $\mathbf{E}\, C_{p,q} = 0$ and $\mathrm{Var}(C_{p,q}) = D_p D_q/(N-1)$. For triples, conditioning on Ω_p makes $C_{p,q}$ and $C_{p,r}$ conditionally independent with zero mean, so $\mathbf{E}(C_{p,q}C_{p,r}) = 0$. Expanding $(E^\circ)^2$ and using orthogonality gives (19). □

12.9 The operator norm and eigenvector localization

Define the normalized off-diagonal covariance matrix H with entries

$$H_{p,q} = \frac{C_{p,q}}{\sqrt{D_p D_q}} \quad (p \neq q), \qquad H_{p,p} = 0.$$

Then $V^\circ = S_{\mathrm{diag}} + v^T H v$ where $v_p = w_p\sqrt{D_p}$ and $|v|^2 = D_w$. Consequently, $V^\circ \le (1 + \|H\|_{\mathrm{op}})\, S_{\mathrm{diag}}$.

N	K	$\|H\|_{\mathrm{op}}$	$\lambda_{\max}$	$\lambda_{\min}$	Rayleigh(v)
256	39	0.824	+0.824	−0.706	0.022
512	67	0.987	+0.987	−0.576	0.031
1024	113	1.088	+1.088	−0.707	0.021
2048	205	0.896	+0.896	−0.735	0.004
4096	380	0.904	+0.904	−0.734	0.002
8192	742	1.210	+1.210	−1.000	0.009

Table 2: Spectral data for the normalized covariance matrix H. The operator norm $\|H\|_{\mathrm{op}}$ appears bounded (fluctuating between 0.8 and 1.2, not growing with N). The Rayleigh quotient $v^T H v/|v|^2 = E^\circ/D_w$ for the logarithmic weight vector v is much smaller and non-monotone with mean ≈ 0.01, explaining why $V^\circ \approx S$. At $N = 8192$, two eigenvalues equal -1 exactly, arising from a three-prime collision cluster with $A_p \in \{1, 2, 3\}$.

The striking gap between $\|H\|_{\mathrm{op}} \approx 1$ and the Rayleigh quotient ≈ 0.01 is explained by *eigenvector localization.* At every computed N, the top eigenvector concentrates on two or three primes with very sparse columns ($A_p \le 4$): these are large primes that share a common marked integer by CRT coincidence. At $N = 8192$ ($\lambda_{\max} = 1.21$), the cluster consists of $p = 14851, 7937, 7559$ with $A_p = 1, 2, 3$ respectively. The logarithmic weight vector $v_p = (\log p)\sqrt{D_p}$ assigns negligible mass to these sparse primes (combined contribution $< 0.5\%$ of $|v|^2$ at every N), so it is nearly orthogonal to the localized extreme eigenvectors. At $N = 4096$, only 3% of $|v|^2$ lies in the top 5 eigenvectors; the largest projection of v (10%) falls on an eigenvalue of magnitude 0.14, deep in the spectral bulk. At $N = 8192$, the projection drops to 0.5%.

This localization has a combinatorial origin: the extreme eigenvalues of H arise from k-fold CRT collisions—integers m marked by $k \ge 3$ distinct large primes. The corresponding eigenvectors span a $(k-1)$-dimensional space of eigenvalue -1 and one direction of eigenvalue $k-1$. The logarithmic weight vector, being smoothly spread across all K active primes, is structurally immune to these localized collision artifacts.

12.10 Palm decorrelation and the cancellation mechanism

The identity (17) decomposes $V^\circ = S_{\mathrm{diag}} + E^\circ$ where the off-diagonal $E^\circ = v^T H v$. The Rayleigh quotient $R_v = v^T H v/|v|^2 = E^\circ/D_w$ admits a transparent conditional-expectation interpretation.

Define the *load excess*

$$\eta_p \;=\; \frac{(Hv)_p}{v_p} \;=\; \frac{\dfrac{1}{A_p}\displaystyle\sum_{m\in\Omega_p}\big(B_{-p}(m)-\mu_{-p}\big)}{w_p\,(1-A_p/N)}, \tag{21}$$

where $B_{-p}(m) = B(m) - w_p X_p(m)$ is the total load excluding prime p, and $\mu_{-p} = \mu - w_p A_p/N$. The quantity η_p measures the average excess load from all *other* primes, conditional on $m \in \Omega_p$, normalized by the centered self-weight. Under exact independence, $\eta_p = 0$ for every p. The Rayleigh quotient is the weighted average

$$R_v \;=\; \sum_p \frac{v_p^2}{D_w}\,\eta_p. \tag{22}$$

Thus $V^\circ \approx S$ says: averaged with the natural variance weights, conditioning on one prime hitting does not bias the load from all other primes. This is the arithmetic content of the observed approximate orthogonality.

Remark 76 (Cancellation, not near-null). The small Rayleigh quotient does *not* mean that v is a near-null vector of H. Computation gives $\|Hv\|/|v| \approx 0.31$ at all tested sizes $N = 512, \ldots, 4096$, while $R_v = v^T Hv/|v|^2$ decreases from 0.031 to 0.002. The cancellation indicator $\|Hv\|/(|v|\,|R_v|)$ grows from 12 to 154: positive and negative spectral contributions to R_v individually exceed 0.13 but nearly cancel. This is the standard trace-zero mechanism: since $\operatorname{tr} H = \sum_j \lambda_j = 0$ and the projections $|\langle u, e_j\rangle|^2$ are approximately uniform ($u = v/|v|$ is delocalized), the identity $R_v = \sum_j \lambda_j(|\langle u, e_j\rangle|^2 - 1/K)$ produces cancellation to order $K^{-1/2}$. The extreme eigenvalues are localized (IPR of top eigenvector ≈ 0.25, effective dimension ≈ 4), so their projections deviate from $1/K$ but contribute little to the weighted sum.

12.11 Dyadic block compression

Partition the active primes into $L = O(\log N)$ dyadic blocks $\mathcal{B}_1, \ldots, \mathcal{B}_L$ according to $\lfloor \log_2 p\rfloor$. For each block, define the restricted weight $s_j = \|v|_{\mathcal{B}_j}\|$ and the unit share $a_j = s_j/|v|$. The *compressed covariance matrix* $\tilde{H} \in \mathbb{R}^{L\times L}$ is

$$\tilde{H}_{i,j} \;=\; \frac{(v^{(i)})^T H\, v^{(j)}}{s_i\, s_j}, \tag{23}$$

where $v^{(j)}$ is v restricted to $\mathcal{B}_j$. Then the Rayleigh quotient factors exactly as

$$R_v \;=\; a^T \tilde{H}\, a. \tag{24}$$

This reduces the $K \times K$ problem to an $L \times L$ one ($L \le 8$) in the single direction of interest.

The compressed norm $\|\tilde{H}\|_{\mathrm{op}}$ controls the Rayleigh quotient via $|R_v| \le \|\tilde{H}\|_{\mathrm{op}}$, and is both numerically accessible (an $L \times L$ eigenvalue problem) and analytically tractable (each entry is a doubly-weighted average of covariances over a dyadic block pair).

The Marchenko–Pastur null model for K independent isotropic columns in $\mathbb{R}^N$ predicts $\|H\|_{\mathrm{op}} \approx 2\sqrt{K/(N-1)} + K/(N-1)$; with $K \sim N/\log N$, this gives $2/\sqrt{\log N} + 1/\log N$, which matches the observed range 0.8–1.2 to within 20–40% (the excess is consistent with sparse localized spikes from low-A_p columns). For the compressed matrix, the same model predicts $\|\tilde{H}\|_{\mathrm{op}} \sim \sqrt{L/N} \sim \sqrt{\log N/N} \to 0$, consistent with the observed decrease.

Remark 77 (Row leverage). The maximum *normalized row leverage* $\rho(n) = \sum_{p:\, n\in\Omega_p} 1/D_p$ is bounded near 1.5 at all tested sizes, dominated by one singleton column ($1/D_p \approx 1$) and one or two doublet columns ($1/D_p \approx 0.5$). No singleton collisions (two singletons at the same m) occur

N	K	L	$\|H\|_{\rm op}$	$\|\tilde H\|_{\rm op}$	R_v	$\|Hv\|/\|v\|$
512	67	6	0.987	0.113	0.031	0.386
1024	113	6	1.088	0.111	0.021	0.302
2048	205	7	0.896	0.097	0.004	0.311
4096	380	7	0.904	0.040	0.002	0.313
8192	742	8	1.210	0.052	0.009	0.291

Table 3: Full operator norm $\|H\|_{\rm op}$ versus compressed operator norm $\|\tilde H\|_{\rm op}$. The full norm is bounded near 1 with a spike to 1.21 at $N = 8192$ (from the first singleton collision); the compressed norm is much smaller, ranging from 0.113 at $N = 512$ to 0.040–0.052 at $N = 4096$–8192. The column $\|Hv\|/|v|$ shows that v is not a near-null vector of H ($\|Hv\|/|v| \approx 0.3$ at all sizes); the small Rayleigh quotient arises from spectral cancellation.

up to $N = 4096$. By the Gram identity $H + I = U^T U$ where U is the $N \times K$ normalized incidence matrix, $\|H\|_{\rm op} + 1 \le \max_n \rho(n) \cdot K$ is a trivial upper bound. A tighter bound via $\max_n \rho(n)$ governs the localized spectral spikes; if $\rho(n) \le C$ uniformly, then $\|H\|_{\rm op} \le C - 1$.

12.12 Gram identity and block aggregates

The compression admits an exact Gram-matrix interpretation. Define the *normalized block aggregate*

$$Y_i(m) \;=\; \frac{1}{s_i} \sum_{p \in \mathcal{B}_i} (\log p) \left(X_p(m) - \frac{A_p}{N} \right), \qquad 1 \le i \le L, \tag{25}$$

where each Y_i is a centered function on I_N with $\sum_m Y_i(m) = 0$. Let Y be the $N \times L$ matrix with columns Y_i. Then

$$\boxed{I_L + \tilde H \;=\; Y^T Y \;\succeq\; 0,} \tag{26}$$

or entrywise, $\tilde H_{i,j} = \langle Y_i, Y_j \rangle - \delta_{ij}$. This is the key structural identity: $I + \tilde H$ is the Gram matrix of L vectors in $\mathbb{R}^N$, so every eigenvalue of $\tilde H$ is at least -1. Only the positive edge requires an arithmetic bound.

The Rayleigh identity (24) now reads

$$R_v \;=\; a^T \tilde H\, a \;=\; \left\| \sum_i a_i Y_i \right\|_2^2 - 1.$$

Thus $R_v = O(1)$ is equivalent to approximate orthonormality of the L aggregate functions Y_i.

Remark 78 (Low-to-high transfer). Let $P = EE^T$ denote orthogonal projection onto the L-dimensional block-smooth subspace. Then $PHu = E\tilde H a$ and

$$\|Hu\|^2 \;=\; \|\tilde H a\|^2 + \|(I - P)HEa\|^2. \tag{27}$$

At $N = 4096$, the data give $\|Hu\| \approx 0.31$ and $\|PHu\| \le \|\tilde H\|_{\rm op} \approx 0.04$, so $\|(I - P)Hu\| \ge 0.307$. Almost all of Hu lies in the within-block oscillatory subspace, not the block-smooth subspace. The covariance operator converts a smooth prime-scale profile into oscillations inside the blocks. This is a strictly sharper diagnosis than spectral cancellation alone.

12.13 Exact random CRT prediction

The independent *fixed-margin* CRT model (Section 12.8) yields exact second-moment formulas for the compressed matrix. Each normalized column satisfies the isotropy identity $\mathbf{E}[\phi_p\phi_p^T] = \frac{1}{N-1}P_0$, where $P_0 = I - \frac{1}{N}\mathbf{1}\mathbf{1}^T$. For distinct primes, $\mathbf{E}[H_{p,q}^2] = \frac{1}{N-1}$.

Proposition 79 (Compressed second moments)**.** *In the independent fixed-margin model,* $\mathbf{E}[\tilde{H}_{i,j}] = 0$ *for all* i, j*, and*

$$\mathbf{E}[\tilde{H}_{i,j}^2] = \frac{1}{N-1} \qquad (i \neq j), \tag{28}$$

$$\mathbf{E}[\tilde{H}_{i,i}^2] = \frac{2}{N-1}(1-\kappa_i), \qquad \kappa_i = \sum_{p\in\mathcal{B}_i} e_i(p)^4. \tag{29}$$

Consequently, the Frobenius norm satisfies

$$\mathbf{E}\|\tilde{H}\|_F^2 = \frac{L^2 + L - 2\sum_i \kappa_i}{N-1} \asymp \frac{L^2}{N}. \tag{30}$$

Proof. For $i \neq j$, disjoint block supports give $\tilde{H}_{i,j} = \sum_{p\in\mathcal{B}_i}\sum_{q\in\mathcal{B}_j} e_i(p)e_j(q)H_{p,q}$. Since different unordered prime pairs have uncorrelated entries (including those sharing one prime, by conditioning), only the diagonal terms survive: $\mathbf{E}[\tilde{H}_{i,j}^2] = \frac{1}{N-1}\sum_p e_i(p)^2 \sum_q e_j(q)^2 = \frac{1}{N-1}$. The diagonal entries involve $\sum_{p<q\in\mathcal{B}_i} e_i(p)^2 e_i(q)^2$, giving the κ_i correction. □

This proves $\|\tilde{H}\|_{\mathrm{op}} \to 0$ *in the random model* whenever $L = o(\sqrt{N})$. The GOE refinement predicts $\|\tilde{H}\|_{\mathrm{op}} \approx 2\sqrt{L/N}$, with the factor 2 being the asymptotic edge constant.

N	L	$\sqrt{L/N}$	$\|\tilde{H}\|_{\mathrm{op}}$	$\|\tilde{H}\|_{\mathrm{op}}/\sqrt{L/N}$
512	6	0.108	0.113	1.04
1024	6	0.077	0.111	1.45
2048	7	0.058	0.097	1.66
4096	7	0.041	0.040	0.97
8192	8	0.031	0.052	1.67

Table 4: The compressed operator norm tracks the random CRT prediction $\sqrt{L/N}$ with ratio $\approx$ 1–1.7.

Remark 80 (Random surrogate comparison). For each N, we generate 20 random palindromic zero sets of the same sizes $|Z_p|$, build the corresponding $\tilde{H}$, and compare $\|\tilde{H}\|_{\mathrm{op}}$. At $N = 2048$: actual = 0.097, random mean = 0.097, z-score = −0.01. At $N = 1024$: actual = 0.111, random mean = 0.130, z-score = −0.91. The actual Apéry matrix is statistically indistinguishable from its random palindromic surrogate.

12.14 AMTD: the precise remaining conjecture

The dispersion target (Hypothesis 66) is *not* a consequence of the Elliott–Halberstam conjecture, the generalized Elliott–Halberstam conjecture, or Barban–Davenport–Halberstam for fixed sequences. Those conjectures concern the discrepancy of a *fixed* arithmetic function in residue classes; here the tested residue function $1_{\mathcal{Z}_p}$ varies with the modulus. This is a fundamental structural distinction, not a technical gap.

We therefore propose:

Hypothesis 81 (Apéry Moving-Target Dispersion (AMTD))**.** *There exists an absolute constant C such that, for every dyadic interval $I_N = (N, 2N]$ and every dyadic prime block $P < p \le 2P$ with $P > \sqrt{N}$,*

$$V^{\circ}(P,N) \;\le\; C\,N^{o(1)}\,S(P,N).$$

The strongest structural sufficient condition is a *short-arc Fourier orthogonality* conjecture:

$$\int_{\mathbb{R}/\mathbb{Z}} \Bigg| \sum_{\substack{P<p\le 2P \\ 1\le a<p \\ x<a/p\le x+1/N}} \frac{(\log p)\,F_p(a)}{p} \Bigg|^2 dx \;\ll\; \frac{N^{o(1)}}{N} \sum_{P<p\le 2P} \sum_{a=1}^{p-1} \Bigg| \frac{(\log p)\,F_p(a)}{p} \Bigg|^2, \tag{31}$$

where $F_p(a) = \sum_{r\in\mathcal{Z}_p} e_p(ar)$. By Gallagher's lemma, (31) implies AMTD block by block.

The classical additive large sieve gives the right-hand side of (31) multiplied by $(1 + P^2/N)$. For $P > \sqrt{N}$, this factor exceeds 2 and grows to $O(N)$ at the top block $P \sim N$. The P^2 barrier is intrinsic to the generic large-sieve proof and cannot be removed by average-spacing refinements alone. A proof of (31) must therefore exploit the special structure of the Apéry Fourier coefficients $F_p(a)$—their palindromic phase constraint ($e_p(a/2)\,F_p(a) \in \mathbb{R}$), their connection to the Sym^2 K3 Hasse polynomial, or the arithmetic of the zero sets $\mathcal{Z}_p$ themselves.

Remark 82 (Partial large-sieve result for small primes)*.* For dyadic blocks with $P \le \sqrt{N}$, the classical additive large sieve gives $V^{\circ}(P,N) \le (N + 4P^2) \sum_{P<p\le 2P} w_p^2 Z(p)/p \le 5\,S(P,N)$. Thus AMTD holds *unconditionally* for primes below $\sqrt{N}$ with constant $C = 5$. The open case is $P > \sqrt{N}$, where the large-sieve factor $1 + 4P^2/N$ exceeds any fixed constant. At the top block $P \sim N$, this factor is $O(N)$, and the generic large sieve provides no useful bound.

Remark 83 (Linnik dispersion route)*.* The most plausible analytic pathway is Linnik's dispersion method: expand the L^2-discrepancy, isolate the diagonal, and seek cancellation in the off-diagonal via reciprocity and bilinear trace-function estimates à la Kowalski–Michel–Sawin [10] and Drappeau [8]. This reduces the problem to a *bounded-complexity representation theorem*: either the Fourier coefficient $a \mapsto \hat{\mathcal{Z}}_p(a) = \sum_{r\in\mathcal{Z}_p} e_p(ar)$ or the phase $m \mapsto e_p(u\,b_m)$ must belong to a uniformly controlled trace-function family. The obstacle is that $b_m \bmod p$ is computed from a nonautonomous recurrence whose complexity grows with p, and no fixed bounded-conductor ℓ-adic sheaf with trace function $A_p(t)$ uniformly in p is currently known. The closest available framework is Forey–Fresán–Kowalski [11], which studies arithmetic Fourier transforms of a fixed perverse sheaf over a fixed finite field; four mismatches remain (growing complexity, horizontal vs. extension-degree aspect, crystalline vs. ℓ-adic, and one- vs. two-characteristic dispersion). A polylogarithmic loss in (31) already suffices for the competition target.

Remark 84 (Palindromic Fourier factoring)*.* For doublet primes in the top block, the exponential sum over hit positions is

$$S(\theta) = \sum_{p\in\mathcal{D}} \big[e(\theta\,m_1(p)) + e(\theta\,m_2(p))\big] = 2\sum_{p\in\mathcal{D}} e\Big(\theta\,\tfrac{3p-1}{2}\Big)\cos(\pi\theta\,h_p)\,,$$

where $h_p = m_2(p) - m_1(p) = p - 1 - 2r_p$ is the doublet gap. By Parseval, $\int_0^1 |S(\theta)|^2\,d\theta = T + F$. The diagonal AMTD requires this integral to be $O(T)$, equivalently that $S(\theta)$ has approximately square-root cancellation on the minor arcs.

The palindromic structure converts each doublet edge into a prime-indexed phase $e(3p\theta/2)$ modulated by the cosine $\cos(\pi\theta h_p)$. The gaps h_p are determined by the first zero r_p of $b_n \bmod p$. A proof of diagonal AMTD via this route would establish cancellation in $S(\theta)$ by exploiting the "pseudorandom" distribution of h_p as p varies.

Remark 85 (Hierarchy of targets). The compression produces a strict hierarchy of increasingly weak—and therefore increasingly tractable—conjectures:

(A) **Full operator bound:** $\|H\|_{\mathrm{op}} = O(1)$. This implies a Bessel inequality for all prime columns and is far stronger than needed.

(B) **Compressed Bessel bound:** $\|\tilde{H}\|_{\mathrm{op}} = O(1)$. This asks for approximate orthonormality of only $L = O(\log N)$ block-aggregate functions, and suffices for the all-index GCD estimate.

(C) **Scalar Rayleigh bound:** $a^T\tilde{H}a = O(1)$. This is the weakest statement that closes the all-index theorem, equivalent to $V^\circ \ll S$. It preserves all useful signs and remains a two-prime problem.

Each implication (A)⇒(B)⇒(C) is strict. The random CRT model predicts $\|\tilde{H}\|_{\mathrm{op}} \asymp \sqrt{L/N} \to 0$ and $|a^T\tilde{H}a| \asymp N^{-1/2}$, both of which are far stronger than (B) and (C).

Remark 86 (Diagonal AMTD via Cauchy–Schwarz). The all-index GCD theorem admits a simpler closing route that avoids cross-block estimates entirely. Define the uncentered block load $F_i(m) = \sum_{p\in\mathcal{B}_i}(\log p)\, g_p(m) = s_i\, Y_i(m)$, so that $V^\circ = \|\sum_i F_i\|_2^2$. If each dyadic block satisfies

$$\|F_i\|_2^2 \;\le\; C\, s_i^2 \qquad \text{(Diagonal AMTD)}, \tag{32}$$

equivalently $1 + \tilde{H}_{ii} \le C$, then by Cauchy–Schwarz,

$$V^\circ \;\le\; \Big(\sum_i \|F_i\|_2\Big)^2 \;\le\; L\sum_i \|F_i\|_2^2 \;\le\; C L\, S.$$

Since $L = O(\log N) = N^{o(1)}$, this gives $V^\circ \le N^{o(1)}\, S$, which suffices for $\max_m |B(m) - \mu| \le \sqrt{V^\circ} = O\big(\sqrt{N}\,(\log N)^{3/2}\big) = o(N/\log N)$. The within-block estimate (32) is a *two-prime* signed dispersion problem—it preserves useful signs, does not require four-prime correlations, and avoids the full operator norm or Frobenius arguments. At the top block $P \sim N$, it reduces to a sparse translated-set collision problem: bounding the additive energy of $\{p + r : r \in \mathcal{Z}_p\}$ for primes p in a dyadic range.

Block-level data at $N = 4096$. The ratio $\|F_i\|_2^2/s_i^2$ for each dyadic block at $N = 4096$:

Block	K_i	$\|F_i\|_2^2/s_i^2$	$\tilde{H}_{ii}$
2^6	2	1.018	+0.018
2^7	13	0.998	−0.002
2^8	17	0.987	−0.013
2^9	30	0.988	−0.012
2^{10}	47	1.015	+0.015
2^{11}	94	1.016	+0.016
2^{12}	177	0.989	−0.011

Maximum ratio: 1.018 (well below any nontrivial constant). The Cauchy loss is $L \cdot \sum_i \|F_i\|_2^2/S = 7.00 = L$, confirming $V^\circ/S = 1.002$ while the Cauchy bound gives $V^\circ \le 7S$. At $N = 8192$ the maximum block ratio remains 1.029 (block 2^{10}) with $L = 8$ blocks. At $N = 32{,}768$ the collision count for top-block doublets is $F = 38$ with $T = 925$, giving $F/T = 0.041$ (decreasing from 0.055 at $N = 8192$), confirming the conjecture across all tested sizes.

Remark 87 (Translation variance). For each pair of distinct primes p, q, averaging over all translates $x \bmod pq$ gives

$$\frac{1}{pq} \sum_{x \bmod pq} |T_{p,q}(x;N)|^2 \;\leq\; \frac{N\,Z(p)^2 Z(q)^2}{pq},$$

where $T_{p,q}(x;L) = \sum_{j=1}^{L} f_p(x+j)\, f_q(x+j)$. For bounded zero counts $Z(p) = O(1)$, this is already at the random scale for a *typical* translate. However, the GCD problem samples the single translate $x = N$ simultaneously for all prime pairs. Summing individual pair bounds absolutely costs $O(N^{3/2})$, which exceeds $S \asymp N(\log N)^2$. The P^2 large-sieve barrier arises because the Fourier support of f_p has $\sim p$ nonzero frequencies (not $Z(p)$), and palindromic symmetry improves only absolute constants.

12.15 Poisson variance formulation

The diagonal AMTD (32) admits a clean probabilistic reformulation. For a dyadic block $\mathcal{B}_i$, define the *column multiplicity function*

$$\lambda_i(m) = \#\{p \in \mathcal{B}_i : m \in \Omega_p\} = \sum_{p \in \mathcal{B}_i} X_p(m), \qquad m \in I_N.$$

Its empirical mean and variance over I_N are

$$\mu_i = \frac{1}{N} \sum_m \lambda_i(m) = \frac{1}{N} \sum_{p \in \mathcal{B}_i} A_p, \qquad V_i = \frac{1}{N} \sum_m (\lambda_i(m) - \mu_i)^2.$$

Expanding the square gives

$$N\,V_i = \sum_{p \in \mathcal{B}_i} D_p + \sum_{\substack{p,q \in \mathcal{B}_i \\ p \neq q}} C_{p,q}.$$

Since all primes in a dyadic block satisfy $\log p \asymp \log P$, one has $\|F_i\|_2^2 \asymp (\log P)^2\, N\, V_i$ and $s_i^2 \asymp (\log P)^2 \sum_p D_p$, so the diagonal AMTD is equivalent (up to asymptotically negligible corrections from the non-constant weight $\log p$) to:

$$V_i \;\leq\; C\,\mu_i \qquad \text{(sub-Poisson variance).} \tag{33}$$

If the indicators $X_p(m)$ were independent Bernoulli variables (with $\Pr[X_p = 1] = A_p/N$), each $\lambda_i(m)$ would be a sum of independent sparse Bernoullis and the variance-to-mean ratio would be ≤ 1. Condition (33) therefore says that the deterministic Apéry indicators behave as if they were *approximately independent* at the block level.

Table 5 records the empirical variance-to-mean ratio for the top dyadic block.

Remark 88 (Collision count). The off-diagonal covariance sum has the exact identity

$$\sum_{p \neq q \in \mathcal{B}_i} C_{p,q} \;=\; \sum_m \lambda_i(m)\big(\lambda_i(m) - 1\big) \;-\; \frac{(\sum_p A_p)^2 - \sum_p A_p^2}{N}.$$

The first term counts pairwise collisions (integers hit by at least two primes in the block). For the top block ($P \sim N$), each prime has $A_p \leq Z(p) \leq 3$ hits, so $\lambda_i(m) \leq 3$ for all m at $N = 8192$. The collision count fluctuates around the Poisson prediction T^2/N, confirming the near-independence of the indicators.

N	K_{top}	μ	V/μ	collisions	Poisson pred.
1024	47	0.086	1.073	14	7.6
2048	94	0.079	0.983	10	12.8
4096	177	0.079	0.989	22	25.6
8192	363	0.080	1.011	60	52.9

Table 5: Column multiplicity statistics for the top dyadic block. $\mu = \frac{1}{N}\sum_p A_p$ is the mean, V/μ is the variance-to-mean ratio (Poisson prediction: 1), "collisions" is $\sum_m \lambda(\lambda-1)$, and "Poisson pred." is T^2/N where $T = \sum_p A_p$.

Remark 89 (Doublet dominance)*.* Palindromic symmetry $b_{p-1-j} \equiv b_j \pmod p$ forces the zero set $\mathcal{Z}_p$ to be closed under $r \mapsto p-1-r$. If $Z(p) > 0$, then $Z(p)$ is typically even, and indeed at the top block, $Z(p) \geq 2$ for all active primes. The distribution at $N = 8192$ is: *doublets* $Z = 2$ (78%), *quadruplets* $Z = 4$ (19%), *sextuplets* $Z = 6$ (3%), *octuplets* $Z = 8$ ($< 1\%$). No singleton primes appear ($Z = 1$ requires $b_{(p-1)/2} \equiv 0 \pmod p$, which is extremely rare for large p).

For a doublet prime with $\mathcal{Z}_p = \{r_p, p-1-r_p\}$, the two hit positions are

$$m_1(p) = p + r_p, \qquad m_2(p) = 2p - 1 - r_p,$$

satisfying $m_1(p) + m_2(p) = 3p - 1$. A collision between two doublet primes p, q requires one of four types:

$$\begin{aligned}
&\text{(A) } m_1(p) = m_1(q) &&\iff && r_q = r_p + (p-q),\\
&\text{(B) } m_1(p) = m_2(q) &&\iff && r_p + r_q = 2q - 1 - p,\\
&\text{(C) } m_2(p) = m_1(q) &&\iff && r_p + r_q = 2p - 1 - q,\\
&\text{(D) } m_2(p) = m_2(q) &&\iff && r_q = r_p + 2(q-p).
\end{aligned}$$

For $q > p$, type (B) is impossible: $m_1(p) = p + r_p < 3p/2 < 3q/2 < 2q - 1 - r_q = m_2(q)$. The collision energy therefore decomposes into three, not four, pieces:

$$F = F_A + F_C + F_D, \tag{34}$$

where $F_A = \sum_m \nu_1(m)(\nu_1(m)-1)$, $F_D = \sum_m \nu_2(m)(\nu_2(m)-1)$, $F_C = 2\sum_m \nu_1(m)\nu_2(m)$, with $\nu_1(m) = \#\{p : m_1(p) = m\}$ and $\nu_2(m) = \#\{p : m_2(p) = m\}$.

Two distinct doublet columns $\{m_1(p), m_2(p)\}$ and $\{m_1(q), m_2(q)\}$ can share at most one element: if both elements coincided, then $m_1(p)+m_2(p) = m_1(q)+m_2(q)$, i.e., $3p-1 = 3q-1$, forcing $p = q$. Thus $J_{p,q} := |\Omega_p \cap \Omega_q| \leq 1$ for distinct doublet primes in the top block. The doublet incidence system is a *simple graph* (no multi-edges), with the ordered collision count $F = \sum_m \lambda(m)(\lambda(m)-1)$. The exact variance identity is

$$V = T + F - T^2/N.$$

Since $T \asymp N/\log N$ and $T^2/N = o(T)$, the diagonal AMTD ($V \leq CT$) is equivalent to the collision bound $F = O(T)$.

Collision decomposition.

N	K	T	T^2/N	F	F_A	F_C	F_D	F/T
1024	47	39	1.5	2	2	0	0	0.051
2048	93	70	2.4	4	2	2	0	0.057
4096	177	140	4.8	0	0	0	0	0.000
8192	363	293	10.5	16	10	4	2	0.055
16384	669	528	17.0	22	16	6	0	0.042
32768	1145	925	26.1	38	18	20	0	0.041

The ratio F/T is consistently small and decreasing, empirically confirming $F = o(T)$. Under the FPI prediction $F \approx 3K^2/N$, one expects $F/T \approx 3/(6.6 \log N)$, giving 0.044 at $N = 32{,}768$, in good agreement with the observed 0.041. At all sizes $\max_m \lambda(m) \le 3$.

Remark 90 (Two-prime divisor formulation). At the top block ($P > N$), each hit $m \in \Omega_p$ implies $p \mid b_m$ by the Lucas congruence. Therefore

$$\lambda_{\mathcal{B}}(m) = \omega_{(N,2N]}(b_m) := \#\{p \text{ prime} : N < p \le 2N,\ p \mid b_m\},$$

and the collision count becomes

$$F = \sum_{m \in I_N} \omega_{(N,2N]}(b_m)\big(\omega_{(N,2N]}(b_m) - 1\big).$$

The diagonal AMTD is therefore equivalent to the following *two-large-prime divisor theorem*: on average over m, the Apéry number b_m has at most $O(1)$ prime divisors in the dyadic range $(N, 2N]$.

This formulation reveals the arithmetic core: the collision bound $F = O(T)$ is an Apéry-specific Barban–Davenport–Halberstam theorem, asking for a level-of-distribution estimate at the second prime moment. A single-prime level-of-distribution (equivalently, $T = O(N/\log N)$) follows from the Poisson conjecture and is not in doubt; the missing step is the *bilinear* independence: the events $p \mid b_m$ and $q \mid b_m$ are approximately uncorrelated as p, q range over a dyadic block.

Vertical vs. horizontal projections. The incidence graph has primes on one side and integers on the other; an edge (p, n) records $p \mid b_n$. Grouping by the integer index n gives the *vertical* prime-divisor count $W_N(n) = \omega_{(N,2N]}(b_n)$, $n \in [1, N]$. Grouping by the hit position $m = n + p$ gives the *horizontal* column multiplicity $\lambda(m)$. These are different projections. By the Lucas congruence $b_{p+r} \equiv 5\, b_r \pmod{p}$ (for $r < p$), the horizontal multiplicity $\lambda(m) = \omega_{(N,m]}(b_m)$—it counts large prime divisors of b_m itself. Crucially, even the pointwise bound $W_N(n) \le 1$ for *every* $n \le N$ does not control $\lambda(m)$: many distinct (n_i, p_i) pairs can satisfy $n_i + p_i = m$ with $p_i \mid b_{n_i}$, all contributing *different* prime divisors of the single integer b_m. The collision bound $F = O(T)$ is therefore a genuinely horizontal, two-prime-divisor assertion about b_m.

Proposition 91 (Conditional collision bound)**.** *Assume the following* Frobenius Pair Independence *hypothesis (FPI): for distinct doublet primes $p, q \in (N, 2N]$, the normalized first zeros r_p/p and r_q/q are asymptotically independent and each approximately uniform on $[0, \frac{1}{2}]$ in the sense that, for any integers $a \in [1, (p-1)/2]$ and $b \in [1, (q-1)/2]$,*

$$\frac{1}{K(K-1)} \#\big\{(p,q) \text{ distinct doublet primes in } (N,2N] : r_p = a,\ r_q = b\big\} \;=\; \big(1 + o(1)\big)\,\frac{4}{pq}, \qquad (35)$$

where K is the number of doublet primes. Then $F = o(T)$ and the all-index GCD theorem $G_n = e^{o(n)}$ holds for all n.

Proof. We bound each piece of the decomposition (34) separately.

Type A. The type-A collision count is

$$F_A = \#\{(p,q) \text{ distinct doublet primes} : r_q = r_p + (p-q)\}.$$

For a fixed ordered pair (p,q), the collision requires $r_q = r_p + (p-q)$. Summing the FPI probability over all valid values of r_p, the conditional probability is

$$\Pr[\text{type A} \mid p,q] \;=\; \sum_{r=1}^{(p-1)/2} \frac{2}{p}\cdot\frac{2}{q}\cdot \mathbf{1}\big[r+(p-q)\in[1,(q-1)/2]\big] \;\le\; \frac{4\,\min\big((p-1)/2,\,(3q-1)/2-p\big)}{pq}.$$

For primes $p,q\in(N,2N]$, this is at most $4/N$ when $|p-q|\le N/2$, and vanishes when $|p-q|>N$. Summing over all $K(K-1)$ ordered pairs:

$$F_A \;\le\; K(K-1)\cdot\frac{4}{N}\cdot(1+o(1)) \;\le\; \frac{4K^2}{N}\cdot(1+o(1)).$$

Since $K \asymp N/\log N$ and $T = 2K + O(K^{1/2})$, we obtain $F_A \le 4T^2/(4N)\cdot(1+o(1)) = T^2/N\cdot(1+o(1)) = o(T)$.

Type C. The collision condition $r_p + r_q = 2p-1-q$ constrains a sum of independent uniforms. Under FPI, $r_p + r_q$ has a trapezoidal distribution on $[2,(p+q-2)/2]$ with maximum density $2/\min(p,q)\le 2/N$. The target $2p-1-q$ lies in the support when $q<2p-1$, which holds for $p>(q+1)/2$. Thus $\Pr[\text{type C}\mid p,q]\le 2/N$, and $F_C \le 2K^2/N\cdot(1+o(1)) = o(T)$.

Type D. The condition $r_q = r_p + 2(q-p)$ has the same structure as type A with gap $2(q-p)$ in place of $p-q$. By the same argument, $F_D \le 4K^2/N\cdot(1+o(1)) = o(T)$.

Combining all three, $F = F_A + F_C + F_D = o(T)$. The diagonal AMTD $\|F_i\|_2^2 \le C\,s_i^2$ follows with $C = 1+o(1)$ at each dyadic block, giving $V^\circ \le (1+o(1))\,L\,S = O(\log N)\,S = N^{o(1)}\,S$. ☐

Remark 92 (Status of the FPI hypothesis). The hypothesis (35) is a pair decorrelation statement for the zero positions across different primes. The marginal equidistribution of r_p/p is supported by the Kummer structure of the reduced generating series (Caruso–Fürnsinn–Vargas-Montoya–Zudilin [7]), which shows that the Galois group of f_α mod p is cyclic of order $(p-1)/\gcd(2,p-1)$. The rank-three Apéry Picard–Fuchs connection is the symmetric square of a rank-two connection, so its connected geometric monodromy is of $\mathrm{Sym}^2\mathrm{SL}_2 \simeq \mathrm{SO}_3$ type.

However, neither the cyclic Kummer group nor the SO_3 monodromy controls the zero *positions* r_p. To see why, observe that the zero-set $\mathcal{Z}_p = \{j<p : p \mid b_j\}$ is a *spectral* zero-set: writing $A_p(t) = \sum_{k=0}^{p-1} b_k t^k \in \mathbb{F}_p[t]$ and $f_p(u) = A_p(g^u)$ for a primitive root g, multiplicative orthogonality gives

$$\sum_{u=0}^{p-2} f_p(u)\, g^{-ju} = -b_j,$$

so $j\in\mathcal{Z}_p$ iff the j-th discrete Mellin coefficient of the Hasse-polynomial value vector vanishes. By contrast, the *roots* of $A_p(t)$ in $\mathbb{F}_p$ are evaluation zeros, controlled by the Kummer factorization $A_p(t) = \Delta(t)^{\varepsilon_p} B_p(t)^2$. The Mellin coefficient index j and the discrete logarithm $\mathrm{ind}_g(x)$ of a root x live in dual spaces; no theorem identifies a coefficient-zero position with an evaluation-zero position.

The pair independence (35) is therefore a genuinely new hypothesis: it concerns the spectral zero process $\mathcal{A}_p(j) = \mathbf{1}_{\{b_j\equiv 0\}}$, not a Frobenius class function or a discrete logarithm. Neither

Chebotarev, Sato–Tate, nor Katz equidistribution (which control traces and evaluation points) applies to this spectral-zero process.

A clean unconditional formulation replaces FPI by the *marked two-characteristic Hasse dispersion* conjecture:

$$\sum_{\substack{N<p,q\le 2N\\ p\ne q}}\ \sum_{m\in I_N} X_p(m)\, X_q(m) \ \ll\ T_N + \frac{T_N^2}{N}, \tag{36}$$

where $X_p(m) = \mathbf{1}_{\{p\le m,\ b_{m-p}\equiv 0 \pmod p\}}$. The left side equals the ordered collision count F_N^{ord}. Via the exponential-sum detection $\mathcal{A}_p(j) = p^{-1}\sum_{a \bmod p} e_p(a\, b_j)$, the conjecture (36) reduces to cancellation in the mixed-modulus sum

$$\sum_{p\ne q}\frac{1}{pq}\ \sideset{}{'}\sum_{\substack{a \bmod p\\ c \bmod q}}\ \sum_{m\in I_N} e_p(a\, b_{m-p})\, e_q(c\, b_{m-q}), \tag{37}$$

where $\sum'$ omits the term $a = c = 0$ (which produces the main term). The inner sum is over a nonautonomous recurrence orbit: $m \mapsto (b_{m-p}, b_{m-q})$. Standard Weil bounds do not apply.

A proof of (36) would require (1) a uniform crystalline Mellin object whose first Hasse coordinate tracks b_j mod p; (2) nondegeneracy of this coordinate on the monodromy torsor; (3) product monodromy for distinct Kummer twists; (4) rare-event anti-concentration at $1/p$ scale; and (5) a *horizontal* large sieve across different characteristics—a five-stage program beyond current technology but strongly supported by computation ($F/T \le 0.041$ at $N = 32{,}768$).

Proposition 93 (Profile decomposition)**.** *For each prime $p \in (N, 2N]$, let $A_p = |\Omega_p \cap I_N|$ and $L_p = 2N - p$. Define the smooth tail profile*

$$\rho(m) = \sum_{\substack{N<p\le 2N\\ p<m}} \frac{A_p}{L_p}, \qquad m \in I_N.$$

Then:

(i) $\sum_{m\in I_N} \rho(m) = T_N$;

(ii) $\|\rho\|_2^2 := \sum_{m\in I_N} \rho(m)^2 \le T_N^2/N$;

(iii) $F_N \le 2\|\rho\|_2^2 + 2\|\lambda-\rho\|_2^2 - T_N \le 2T_N^2/N + 2\|\lambda-\rho\|_2^2$.

In particular, the residual dispersion estimate $\|\lambda-\rho\|_2^2 \ll T_N$ would imply $F_N = O(T_N)$.

Proof. Part (i): $\sum_m \rho(m) = \sum_p (A_p/L_p)\cdot L_p = \sum_p A_p = T_N$. For (ii), write $\|\rho\|_2^2 = \sum_{p,q}(A_pA_q)/(L_pL_q)\cdot(2N - \max(p,q))$. By the PNT approximation $\rho(m) \approx (2/\log N)\log(N/(2N-m))$ and the integral $\int_0^1 (\log 1/t)^2\, dt = 2$, one obtains $\|\rho\|_2^2 \sim c\,T_N^2/N$ for an effective constant c. Numerically, $\|\rho\|_2^2/(T_N^2/N)$ stabilizes near 1.25 for $N = 512, \ldots, 32{,}768$ (Table 6). Part (iii) follows from $\|a+b\|^2 \le 2\|a\|^2 + 2\|b\|^2$ applied to $\lambda = \rho + \eta$. □

Remark 94 (Orthogonal decomposition). Setting $\eta = \lambda - \rho$, the identity $T + F = \|\lambda\|_2^2 = \|\rho\|_2^2 + 2\langle\rho,\eta\rangle + \|\eta\|_2^2$ gives

$$F = \|\rho\|_2^2 + 2\langle\rho,\eta\rangle + (\|\eta\|_2^2 - T). \tag{38}$$

Under the independent profile model (each $X_p(m)$ is an independent Bernoulli with mean $\pi_p(m) = A_p/L_p$ when $p < m$), the three terms in (38) have the following expected behavior:

- $\|\rho\|_2^2 = \Theta(T^2/N)$: the deterministic profile energy, tending to 0 relative to T.
- $\|\eta\|_2^2 \approx T$: the shot-noise variance of T independent Bernoulli hits.
- $\langle \rho, \eta \rangle \approx 0$: the profile and residual are decorrelated.

Hence $F \approx \|\rho\|_2^2 \approx T^2/N = o(T)$. Computation (Table 6) confirms this decomposition: $\|\eta\|_2^2/T$ is stable near 1.0 (matching palindromic random surrogates), while $F \approx \|\rho\|_2^2$ to within 1% at $N = 16{,}384$ and within 8% at $N = 32{,}768$. The cross-term $\langle \rho, \eta \rangle$ is $O(1)$ at every tested scale (e.g. -0.03 at $N = 16{,}384$ and 0.40 at $N = 32{,}768$, versus $T = 664$ and 1182). Moreover, the cross-term has a *deterministic* bound: $|\langle \rho, \eta \rangle| \leq \rho_{\max} \cdot T$ where $\rho_{\max} = \max_m \rho(m) = \sum_p A_p/(2N - p)$. Since $\rho_{\max}/(T/N)$ stabilizes near 1.8 (Table 6), $|\langle \rho, \eta \rangle| \leq 1.8\, T^2/N = o(T)$. The unconditional collision theorem $F = O(T)$ is therefore equivalent to the shot-noise estimate $\|\eta\|_2^2 = O(T)$, which is the residualized AMTD (Hypothesis 81).

Table 6: Profile decomposition diagnostics for the full hit measure μ_N. All norms over $I_N = (N, 2N]$. "Surr" denotes the mean of $\|\eta\|_2^2/T$ over 10 palindromic random surrogates.

N	T	F	$\|\rho\|_2^2$	$\|\rho\|_2^2/(T^2/N)$	$\|\eta\|_2^2/T$	Surr
512	21	2	1.1	1.28	1.05	0.98
1024	50	2	3.0	1.22	0.98	0.99
2048	86	4	4.5	1.26	0.99	1.01
4096	172	2	9.4	1.31	0.96	1.00
8192	354	20	19.6	1.28	1.00	1.00
16384	664	34	33.7	1.25	1.00	1.01
32768	1182	58	53.8	1.26	1.00	—

12.16 Gap polynomial sparsity

For the Apéry recurrence, the gap polynomial N_h (whose roots in $\mathbb{F}_p$ are necessary for a simultaneous zero pair $b_x \equiv b_{x+h} \equiv 0$) has $r_p(h) = \#\{x \in \mathbb{F}_p : N_h(x) = 0\} \leq 3(h-1)$, giving $\sum_{h<H} r_p(h) \leq 3H^2/2$ and hence $Z(p) \leq Cp^{2/3}$. However, the actual root counts are drastically smaller: for all primes $p \leq 10{,}000$ with $Z(p) \geq 2$ and all gaps $h \leq 100$, $\sum_{h \leq 100} r_p(h) \leq 1$, with most primes having $\sum = 0$ (the zero pairs are spaced more than 100 apart).

Hypothesis 95 (Linear root sum)**.** *$\sum_{1 \leq h < H} r_p(h) \ll H$ for every prime p and every $H \geq 1$.*

This would improve the zero-count to $Z(p) \ll p^{1/2}$, strengthening the exponent in Proposition 67 from $N^{5/6}$ to $N^{3/4}$. The heuristic justification is that N_h has degree $3(h-1)$ and (conjecturally) Galois group close to $S_{3(h-1)}$ over $\mathbb{Q}$, so that by the Chebotarev density theorem $\mathbb{E}[r_p(h)] = 1$ for each h.

12.17 The exact random model

The palindromic identity $b_{p-1-j} \equiv b_j \pmod{p}$ forces the zero-set to consist of reflected pairs. Let V_p^+ denote the space of inversion-invariant functions $f\colon \mathbb{F}_p^\times \to \mathbb{F}_p$, i.e. $f(t) = f(t^{-1})$. The algebraic discrete Fourier transform $\hat{f}(j) = \sum_{t \in \mathbb{F}_p^\times} f(t)\, t^{-j}$ maps V_p^+ isomorphically onto the subspace of sequences satisfying $\hat{f}(j) = \hat{f}(-j)$.

Proposition 96 (Random inversion-invariant model)**.** *Choose f uniformly from V_p^+. Then the number K of noncentral reflected pairs $\{j, p-1-j\}$ with $\hat{f}(j) = 0$ satisfies*

$$K \;\sim\; \mathrm{Binomial}\Big(\frac{p-3}{2},\; \frac{1}{p}\Big) \;\xrightarrow{p\to\infty}\; \mathrm{Poisson}(1/2).$$

Proof. The spectral coefficients of distinct reflected pairs are independent uniform elements of $\mathbb{F}_p$ (the DFT is invertible), so each vanishes with probability $1/p$. $\square$

The Apéry Hasse polynomial $H_p(t) = \sum b_k t^k$ is a specific element of V_p^+ with additional algebraic structure ($H_p = \Delta^\varepsilon B_p^2$). The conjecture that $Z(p) \sim 2\,\mathrm{Poisson}(1/2)$ is equivalent to asserting that H_p behaves, for the spectral-zero statistic, like a generic element of V_p^+.

The square structure can be incorporated directly. Let $V_p^- = \{g\colon \mathbb{F}_p^\times \to \mathbb{F}_p : g(t^{-1}) = -g(t)\}$ denote the anti-invariant subspace, with $\dim V_p^- = (p-3)/2$. For uniform $g \in V_p^-$, the function $f = g^2$ lies in V_p^+ and vanishes at the two fixed points $t = \pm 1$.

Proposition 97 (Random anti-invariant square model)**.** *Choose g uniformly from V_p^- and put $f = g^2$. For any fixed distinct noncentral reflected pairs represented by $j_1, \dots, j_r$,*

$$\Pr\big(\hat{f}(j_1) = \cdots = \hat{f}(j_r) = 0\big) \;=\; p^{-r} + O_r(p^{-c_r p})$$

for some $c_r > 0$. In particular, the pair-zero count $K^{\mathrm{sq}} \Rightarrow \mathrm{Poisson}(1/2)$.

Proof. Write $u_i = g(x_i)$ for orbit representatives. The spectral coefficient $\hat{f}(j) = \sum_i u_i^2 (x_i^j + x_i^{-j})$ is a diagonal quadratic form $Q_\lambda(u)$ in the independent uniform variables u_i. For $\lambda \neq 0$, the physical function $h_\lambda(t) = \sum_a \lambda_a (t^{j_a} + t^{-j_a})$ has Mellin support $\leq 2r$, so by the finite-field Mattson–Solomon uncertainty inequality, at least $(p-1)/(4r) - 1$ orbit values are nonzero, each contributing a quadratic Gauss sum of absolute value $p^{-1/2}$. The factorial-moment method then gives the result. $\square$

Multiplication by $\Delta(t)^\varepsilon$ affects only $O(1)$ orbits and does not change the limiting law. Thus the Sym^2 squareness of the Apéry Hasse polynomial is fully compatible with Poisson(1/2) in any random model where the square root B_p behaves generically.

Proposition 98 (Quadratic rank and both halves of the spectrum)**.** *Let $B(t)$ be a uniformly random normalized reciprocal polynomial of degree $d = (p-1-2\varepsilon)/2$ over $\mathbb{F}_p$, with free variables $x_1, \dots, x_m$ where $m = \lfloor d/2 \rfloor$. Put $H = \Delta^\varepsilon B^2$ and split the noncentral reflected-pair representatives into a* triangular *range $\mathcal{J}_{\mathrm{tri}}$ ($|\mathcal{J}_{\mathrm{tri}}| = m$) and a* nonlinear *range $\mathcal{J}_{\mathrm{nl}}$ ($|\mathcal{J}_{\mathrm{nl}}| = p/4 + O(1)$).*

(i) *In the triangular range, H_j is an affine-linear function of $(x_1, \dots, x_m)$ with a unitriangular Jacobian. Hence $K_{\mathrm{tri}} = \#\{j \in \mathcal{J}_{\mathrm{tri}} : H_j = 0\} \Rightarrow \mathrm{Poisson}(1/4)$.*

(ii) *In the nonlinear range, each H_j is an affine quadratic form in $(x_1, \dots, x_m)$ whose quadratic part has rank $\geq m/2 - O(1) \gg p$. By the Gauss-sum bound for affine quadratic forms of rank R: $\Pr(F(x) = 0) = 1/p + O(p^{-R/2})$. The pencil-rank estimate from Proposition 97 gives $K_{\mathrm{nl}} \Rightarrow \mathrm{Poisson}(1/4)$.*

(iii) *Without requiring independence between the two halves,*

$$\mathbf{E}\, Z(p) \;=\; 2\,\mathbf{E}\, K \;=\; 2\big(\mathbf{E}\, K_{\mathrm{tri}} + \mathbf{E}\, K_{\mathrm{nl}}\big) \;=\; 2\Big(\frac{1}{4} + \frac{1}{4}\Big) + o(1) \;=\; 1 + o(1).$$

If the mixed pencil-rank condition holds—i.e., every nonzero $\mathbb{F}_p$-linear combination of $H_{j_1}, \dots, H_{j_s}$ with at least one $j_\nu \in \mathcal{J}_{\rm nl}$ has quadratic rank $\gg p/s$—then the full pair-zero count $K = K_{\rm tri} + K_{\rm nl} \Rightarrow$ Poisson(1/2)*.*

Remark 99 (Separation of random model from deterministic problem). Proposition 98 applies to the random reciprocal-polynomial model, not to the actual Apéry sequence. The actual B_p supplies one deterministic coefficient vector for each prime; converting the random-model statement into a theorem about $\sum_{p \le x} Z(p)$ requires an arithmetic anti-concentration theorem asserting that these specific vectors avoid the coefficient quadrics with approximately random frequency. No such theorem is currently known.

12.18 Shift-correlation computation

Define the shift-correlation exponential sum

$$C_p(a) \; = \; \sum_{r=0}^{p-2} e_p(a\,(b_{r+1} - b_r))\,, \qquad a \neq 0.$$

A nontrivial bound $|C_p(a)| \ll p^{1-\eta}$ is the missing input for a Bombieri–Fouvry-type dispersion argument (cf. Section 12.4).

For all 666 primes $p \le 5000$, we computed $\max_{a \neq 0} |C_p(a)|/\sqrt{p}$. The results:

- $\max_a |C_p(a)|/\sqrt{p}$ has mean 3.35 and maximum 5.07;
- $\max_a |C_p(a)|/\sqrt{p \log p}$ has mean 1.23 and maximum 1.80;
- the distribution of $\max/\sqrt{p}$ matches the Gumbel law expected for the maximum of p independent unit-variance Gaussians: $\max \approx \sqrt{2 \log p}$.

For shifts $s = 2, 3, 5$, the same behavior holds. The histogram of the differences $b_{r+1} - b_r \pmod p$ has std/mean ≈ 1 and the fraction of zero-bins is $\approx 1/e$, precisely the Poisson(1) model.

Conclusion: the Apéry sequence $b_r \bmod p$ is computationally indistinguishable from a random function $\{0, \dots, p-1\} \to \mathbb{F}_p$ in its additive exponential-sum statistics. This provides strong computational evidence for the shift-correlation bound required by the dispersion approach.

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